SOS degree lower bound for exact planted clique

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Editorial note. Faithful editable TeX collection of formal Theorem1.4 and the complete author source-local exact-clique lower-bound chain: exact model/parameters; required Section3 pseudoexpectation construction; Section4 homogeneity/concentration proofs and graphical-matrix definitions; Section5.1 Johnson calculation and Section5.3 full recursive factorization/matching/error proof; all Section6 Hadamard/Euler, exact factorization, factorial-Hankel structural positivity, pseudorandom/rest norm bounds and final PSD proof. Repeated statements are not duplicated. Standard cited prior norm/separator facts are exact prerequisites, not recursively re-proved. Optional non-exact mod-order by-product Section5.2/AppendixA and failed example3.9 are not targets or needed proof; the necessary Fact5.7 is retained. All author mathematical expressions/conditions remain uncorrected; no new proof or hypotheses. Complete published bibliography retained. Only official CC BY4.0 content used.

Allowed standard prerequisites (editorial).

- Graphical-matrix norm theorem. Exact author Theorem4.8[24,5], including its parameter bounds and stated moreover derivation, retained.
- Min-separator lattice and standard graph separation. Exact author Fact5.7[11], Menger theorem and credited prior Lemma5.23[5] retained; the author own matching/factorization proofs remain complete.
- Johnson schemes and SOS pseudoexpectation duality. Author model Definitions1.1/1.2, exact Johnson Fact5.3[9] and all own Johnson calculations retained.

Author exact-clique model and SOS conventions

The SOS proof system, also known as the Positivstellensatz system [14], works with polynomials over $\mathbb{R}$. Given polynomial equalities (axioms) $f_1(\mathbf{x}) = 0, \ldots, f_k(\mathbf{x}) = 0$ on $\mathbf{x} = (x_1, \ldots, x_n)$, a proof (a refutation of the existence of a solution) is

$$-1 = \sum_{i=1}^{k} f_i q_i + \sum_j r_j^2 \quad \text{in } \mathbb{R}[x_1, \ldots, x_n],$$

where $q_1, \ldots, q_m$ and $r_1, \ldots$ are arbitrary polynomials on $x_1, \ldots, x_n$ over $\mathbb{R}$. Under certain conditions, in particular when all variables are boolean ($x_i^2 = x_i$), such an refutation always exists if the axioms are contradictory. The degree-d SOS proof system adds the limitation

$$\max_{i,j}\{\deg(f_i) + \deg(q_i), 2\deg(r_j)\} \leq d.$$

Definition 1.1. Given an n-vertex simple graph G and a number ω, the Clique Problem for degree-d SOS proof system has the following axioms:

$$\begin{aligned} \text{(Boolean)} \quad & x_i^2 = x_i && \forall i \in [n], \\ \text{(Clique)} \quad & x_i x_j = 0 && \forall \{i,j\} \text{ non-edge}, \\ \text{(Size)} \quad & x_1 + \cdots + x_n = \omega. && \end{aligned} \tag{1.1}$$

To confirm no ω-clique exists is to give an SOS refutation of the above. The SOS system has duality: to show degree lower bound it suffices to consider pseudo-expectation and the resulting moment matrix.[1] With boolean variables this can be demonstrated on multi-linear polynomials. Let $X^{\leq a} = \{x_S : S \subseteq [n], |S| \leq a\}$ for any a.

Definition 1.2. A degree-d pseudo-expectation for the Clique Problem on G is a map $\widetilde{\mathbb{E}} : X^d \to \mathbb{R}$ satisfying the following four constraints when extended by $\mathbb{R}$-linearity:

$$\text{(Default)} \quad \widetilde{\mathbb{E}} x_\emptyset = 1, \tag{1.2}$$

$$\text{(Clique)} \quad \widetilde{\mathbb{E}} x_S = 0 \qquad \forall S : |S| \leq d,\ G|_S \text{ non-clique}, \tag{1.3}$$

$$\text{(Size)} \quad \widetilde{\mathbb{E}}((x_1 + \cdots + x_n)x_S) = \omega \widetilde{\mathbb{E}} x_S \quad \forall S : |S| \leq d-1. \tag{1.4}$$

In (1.4), $x_A x_B := x_{A \cup B}$. Define the moment matrix M to be the $\binom{[n]}{\leq d/2} \times \binom{[n]}{\leq d/2}$ matrix with expression $M(A,B) = \widetilde{\mathbb{E}} x_{A \cup B}$ for all $|A|, |B| \leq d/2$.[2] The last constraint is:

$$M \text{ is positive semi-definite.} \tag{1.5}$$

If a degree-d pseudo-expectation exists then there is no degree-d SOS refutation. Exact Clique is Definition1.1. The Non-Exact Clique relaxation replaces the Size Constraints by one weaker inequality $\widetilde{\mathbb{E}}(x_1 + \cdots + x_n) \geq \omega$. There is no planted clique in the problem's formulation; traditionally it is still called the planted clique problem due to the algorithmic motivation.

Formal author main result

Theorem 1.4. Let $\epsilon > 0$ be any parameter, $\omega = n^{1/2-\epsilon}$. W.p. $> 1 - n^{-4\log n}$ over $G \sim G(n, \frac{1}{2})$, any SOS refutation of Exact Clique requires degree at least $\epsilon' \log n / \log\log n$, where $\epsilon' = \min\{\epsilon^2, 1/40^2\}/2000$.

[1] The author uses the name for simplicity. More cautiously, it should be called the pseudo-moment matrix.

[2] d is always assumed to be even.

Author notation, parameter regime and Section 3

I, J, A, B, S denote vertex-sets and T edge-sets. $E(S) := \binom{S}{2}$. G denotes a simple graph on vertex-set $[n]$. "$T \subseteq E([n])$" is omitted in summation when there is no confusion. $y(n) = O(x(n))$ means there is some absolute constant c such that $y(n) \leq cx(n)$ for all n. Throughout the paper,

$$\epsilon = \text{any positive parameter (wlog } \epsilon < 1/40), \quad \omega = n^{1/2-4\epsilon}, \quad \tau = \frac{\epsilon}{200}\frac{\log n}{\log\log n}, \quad d = \frac{\epsilon}{100}\tau.$$

3.1 Non-exact case: a new perspective

Given a graph G we can think of a degree-d pseudo-expectation as assigning a number $\mathbb{E}x_S$ to each subset $S \subseteq [n]$ of size $\leq d$, so that the resulting vector $\widetilde{\mathbb{E}}x$ looks indistinguishable to the expectation resulted from the case when a random-ω clique is planted, from the view of degree-d SOS. To make up such an assignment we first go beyond to slightly larger subsets of size τ. We define an "approximate distribution" on size $\leq \tau$-cliques in G then use it to generate pseudo-expectation on all size $\leq d$-subsets.

3.1.1 ζ-function and Möbius inversion

Given n-vertex graph G, let $p(G) \in \mathbb{R}^{2^{[n]}}$ be the max-clique-indicator vector, then $q(G) := \zeta p(G)$ is a vector supported exactly on all cliques in G, where ζ is the $2^{[n]} \times 2^{[n]}$ matrix

$$\zeta(A, B) = 1 \text{ iff } A \subseteq B, \qquad \forall A, B \subseteq [n]. \tag{3.1}$$

In particular, if G itself is a single clique then $q(G)$ is the clique-indicator. $\zeta_{a,b}$ denotes the submatrix of ζ on rows $\binom{[n]}{\leq a}$ and columns $\binom{[n]}{\leq b}$; use similar notation on all related vectors. Consider the plant-situation where G is indeed a single random clique. Suppose its distribution is represented by a plant-distribution vector $p_{\text{plant}} \in \mathbb{R}^{2^{[n]}}$. Let the output-expectation q_{out} be indicator-vector of cliques in G in expectation. Then

$$q_{\text{out}} = \zeta p_{\text{plant}}. \tag{3.2}$$

We call such a pair $(p_{\text{plant}}, q_{\text{out}})$ a plant-setting.

Definition 3.1 (Two plant-settings). The exact plant-setting (p_0, q_0) is:

$$p_0(S) = \binom{n}{\omega}^{-1} \text{ if } |S| = \omega \text{ and 0 otherwise}, \tag{3.3}$$

and

$$q_0(S) = (\zeta p_0)(S) = \frac{\binom{n-|S|}{\omega-|S|}}{\binom{n}{\omega}}. \tag{3.4}$$

I.e. a random size-ω subset is chosen to be the planted clique. The independent plant-setting (p_1, q_1) is:

$$p_1(S) = (\omega/n)^{|S|}(1-\omega/n)^{n-|S|} \quad \forall S \subseteq [n], \tag{3.5}$$

and

$$q_1(S) = (\zeta p_1)(S) = (\omega/n)^{|S|}. \tag{3.6}$$

I.e. any vertex is included in the planted clique w.p. ω/n independently. Thus ζ reveals the basic linear relations between $(p_{\text{plant}}, q_{\text{out}})$. It is upper-triangular (row- and column-indices ordered in a size-ascending way), invertible, and the inverse is the Möbius inversion matrix:

$$\zeta^{-1}(A, B) = (-1)^{|B\setminus A|} \text{ if } A \subseteq B, \text{ and 0 otherwise.}$$

Note $(\zeta_{a,a})^{-1} = (\zeta^{-1})_{a,a}$ for all $a \le n$. Moreover, if let the pseudo-expectation be defined as $\widetilde{\mathbb{E}}x = p \in \mathbb{R}^{2^{[n]}}$ for some vector p, then the "full" $2^{[n]} \times 2^{[n]}$ moment matrix is

$$M_{\text{SOS}} = \zeta \operatorname{diag}(p) \zeta^\top. \tag{3.7}$$

In particular, if p is a nonnegative vector then M_{SOS} is immediately PSD.

3.1.2 The non-exact pseudo-expectation

Given any G, first construct a degree-τ "approximate plant-distribution" $p_\tau(G)$, which simulates the plant-distribution (Definition3.1) in the sense that they give similar output-expectations. Require $p_\tau(G)$ to be supported on size $\le \tau$-cliques in G. Then take $\widetilde{\mathbb{E}}x = \zeta_{d,\tau} p_\tau(G)$ so the result inherits the linear structure posed by ζ. From approximation it seems taking $\zeta^{-1}_{\tau,\tau}(q_1)_\tau$ would suffice, while to make it supported on cliques, same as in [13] add a clique-indicator factor:

$$p_\tau(G)(S) = 2^{|\binom{S}{2}|} \mathrm{Cl}_S(G) \left(\zeta^{-1}_{\tau,\tau}(q_1)_\tau\right)(S) \quad \forall S \subseteq [n],\ |S| \le \tau. \tag{3.8}$$

Here $\mathrm{Cl}_S(\cdot)$ is the clique indicator function and $2^{|\binom{S}{2}|}$ is for re-normalization.

Definition 3.2. $\forall S \subseteq [n]$, the normalized clique-indicator is function

$$\widetilde{\mathrm{Cl}}_S(G) := 2^{|\binom{S}{2}|} \mathrm{Cl}_S(G). \tag{3.9}$$

$\widetilde{\mathrm{Cl}}(G)$ denotes the column vector over a family of S's, always clear from context.

Definition 3.3. The non-exact pseudo-expectation is

$$\widetilde{\mathbb{E}}_{\text{nonexact}} = \zeta_{d,\tau} p_\tau(G) = \zeta_{d,\tau} \left(\widetilde{\mathrm{Cl}}(G) \circ \zeta^{-1}_{\tau,\tau}\right)(q_1)_\tau \in \mathbb{R}^{\binom{[n]}{\le d}}, \tag{3.10}$$

where $\circ$ is Hadamard product.[3] In short, this refines the construction in [13] by one step: factor through size-τ subsets (in the only non-trivial way) so the size-d output inherits linear relations posed by ζ. The resulting moment matrix is

$$M_{\text{nonexact}}(G) = \zeta_{d/2,\tau} \operatorname{diag}(p_\tau(G)) (\zeta_{d/2,\tau})^\top, \tag{3.11}$$

similarly as (3.7).

Remark 3.4. If $p_\tau(G)$ were nonnegative then PSDness would be immediate. Alas, this is not true by computation. PSDness could still possibly hold because $\zeta_{d/2,\tau}$ in (3.11) is degenerate.

[3] In general $(M_1 \circ M_2)M_3 \ne M_1 \circ (M_2 M_3)$, but they are equal if M_1 is a column vector.

Lemma 3.5 (Theorem1.5(1)). For all $S \subseteq [n]$ with $|S| \le d$,

$$\widetilde{\mathbb{E}}_{\text{nonexact}} x_S = \sum_{T:|V(T)\cup S|\le\tau} (\omega/n)^{|V(T)\cup S|}\chi_T. \tag{3.12}$$

Proof. Note $\widetilde{\mathrm{Cl}}_S = \sum_{T\subseteq E(S)} \chi_T$ for all S. Now for S, S' with appropriate size bound,

$$\left(\widetilde{\mathrm{Cl}} \circ \zeta_{\tau,\tau}^{-1}\right)(S,S') = \begin{cases} \sum_{T\in E(S)} \chi_T(-1)^{|S'\setminus S|}, & S \subseteq S', \\ 0, & \text{o.w.} \end{cases}$$

Then

$$\begin{aligned}
&\left(\zeta_{d,\tau}(\widetilde{\mathrm{Cl}} \circ \zeta_{\tau,\tau}^{-1})\right)(S,S') \\
&= \sum_{S'':S\subseteq S''\subseteq S'} \left(\sum_{T\subseteq E(S'')} \chi_T(-1)^{|S'\setminus S''|} \right) \\
&= \sum_{T:V(T)\cup S\subseteq S'} \chi_T \left(\sum_{S'':V(T)\cup S\subseteq S''\subseteq S'} (-1)^{|S'\setminus S''|} \right) \\
&= \sum_{T:V(T)\cup S\subseteq S'} \chi_T \delta_{S'=V(T)\cup S} \\
&= \sum_{T:V(T)\cup S=S'} \chi_T.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\widetilde{\mathbb{E}}_{\text{nonexact}} x_S &= \left(\zeta_{d,\tau}(\widetilde{\mathrm{Cl}} \circ \zeta_{\tau,\tau}^{-1})(q_1)_\tau\right)(S) \\
&= \sum_{S':|S'|\le\tau} \left(\sum_{T:V(T)\cup S=S'} \chi_T (\omega/n)^{|S'|} \right) \\
&= \sum_{T:|V(T)\cup S|\le\tau} \chi_T (\omega/n)^{|V(T)\cup S|},
\end{aligned}$$

for all S with $|S| \le d$. □

3.2 The exact case

In this subsection, we give a generic way to generate possible pseudo-expectations that satisfy Size Constraints (1.4). Define $\widetilde{\mathbb{E}} x_S$ in a top-down fashion: fix it for all $|S| = d$ first, then recursively set

$$\widetilde{\mathbb{E}} x_S \leftarrow \frac{1}{\omega - |S|} \sum_{i\notin S} \widetilde{\mathbb{E}} x_{S\cup\{i\}} \tag{3.13}$$

for smaller-sized S's. Denote by $\widetilde{\mathbb{E}}_d x$ the assignments for $|S| = d$.

Definition 3.6. The d-filtration matrix $\mathrm{Fil}_{d,=d}$, of dimension $\binom{[n]}{\le d} \times \binom{[n]}{d}$, is:

$$\mathrm{Fil}_{d,=d}(A,B) = \begin{cases} \binom{\omega-|A|}{d-|A|}^{-1}, & A \subseteq B \ (|B| = d), \\ 0, & \text{otherwise.} \end{cases} \tag{3.14}$$

Definition 3.7. Given vector $\widetilde{\mathbb{E}}_d x$ assigning a value to each d-subset $S \subseteq [n]$, the exact pseudo-expectation it generates is

$$\widetilde{\mathbb{E}} x := \mathrm{Fil}_{d,=d} \widetilde{\mathbb{E}}_d x. \tag{3.15}$$

Lemma 3.8. The pseudo-expectation in Definition3.7 satisfies the Size Constraints (1.4), regardless of the choice of $\widetilde{\mathbb{E}}_d x$.

Proof. For any $S \in \binom{[n]}{<d}$, take $v_S \in \mathbb{R}^{\binom{[n]}{\leq d}}$,

$$v_S(S') = \begin{cases} \omega - |S|, & S' = S, \\ -1, & S' \supseteq S,\ |S' \setminus S| = 1, \\ 0, & \text{otherwise.} \end{cases}$$

It suffices to show $v_S^\top \mathrm{Fil}_{d,=d} = 0$. But this is a direct check. □

The generated $\widetilde{\mathbb{E}}$ should further satisfy Clique, PSDness and Default Constraints. Default is not a problem as long as $\widetilde{\mathbb{E}} x_\emptyset > 0$, since we can rescale everything by $(\widetilde{\mathbb{E}} x_\emptyset)^{-1}$ without affecting other constraints.

3.3 The exact pseudo-expectation

Take the pseudo-expectation for $|S| = d$ in the form

$$\widetilde{\mathbb{E}} x_S = \sum_{T : |V(T) \cup S| \leq \tau} \chi_T F(|V(T) \cup S|),$$

for a function F called a d-generating function. For general $|S| \leq d$, (3.14) gives, putting $u := d - |S|$,

$$\widetilde{\mathbb{E}} x_S = \frac{1}{\binom{\omega - d + u}{u}} \sum_{T : |V(T) \cup S| \leq \tau} \chi_T \left[\sum_{c=0}^{u} \binom{|V(T) \cup S| - d + u}{c} \cdot \binom{n - |V(T) \cup S|}{u - c} F(|V(T) \cup S| + u - c) \right]. \tag{3.16}$$

Lemma 3.10. Any exact pseudo-expectation generated by (3.16) satisfies Clique and Size Constraints (1.3),(1.4).

Proof. For Clique Constraints, note (3.16) only depends on $|V(T) \cup S|$, so by grouping terms $\widetilde{\mathbb{E}} x_S = \sum_{T : |V(T) \cup I \cup J| \leq \tau} M(I, J; T) \chi_T$ factors through $\widetilde{\mathrm{Cl}}_{I \cup J} = \sum_{T \subseteq E(I \cup J)} \chi_T$. I.e., $M(I, J)(G) = 0$ if $\widetilde{\mathrm{Cl}}_{I \cup J}(G) = 0$. It satisfies Size Constraints by Lemma3.8. □

Definition 3.11 (Exact d-generating function).

$$F(x) := \frac{(x + 8\tau^2)!}{(8\tau^2)!} (\omega / n)^x.$$

Remark 3.12. The design of F, especially its first factor, is technical and the ultimate goal is to make the resulting M positive. The numerator $(x+8\tau^2)!$ will be used in Proposition6.28, where $8\tau^2$ can be replaced by larger polynomials in τ. The $(8\tau^2)!$ denominator is just for convenience (Remark3.14).

Definition 3.13. The exact moment matrix $\widetilde{M}$ is

$$\widetilde{M}(A,B) = \sum_{T:|V(T)\cup A\cup B|\le\tau} \widetilde{M}(A,B;T)\chi_T \quad \forall |A|,|B| \le d/2,$$

where, putting $u = d - |A\cup B|$,

$$\begin{aligned}\widetilde{M}(A,B;T) = \frac{1}{\binom{\omega-d+u}{u}} \Bigg[\sum_{c=0}^{u} &\binom{|V(T)\cup A\cup B| - (d-u)}{c} \\ &\cdot \binom{n-|V(T)\cup A\cup B|}{u-c} \\ &\cdot \frac{(|V(T)\cup A\cup B| + u - c + 8\tau^2)!}{(8\tau^2)!} (\omega/n)^{|(V(T)\cup A\cup B)|+u-c} \Bigg].\end{aligned} \tag{3.17}$$

Author Section 4: preparations

Fourier notation. χ_T is the Fourier character for the edge-set T in the random graph. The author's Fourier orthonormal basis in Equation(7.1) is $\chi_T = \prod_{e\in T}(x_e - (2p-1))/(2\sqrt{p(1-p)})$ for $G(n,p)$; at $p = 1/2$ this is the product of the independent signed edge indicators. This is the standard Fourier convention used throughout the paper.

Remark 3.14 (continued). In (3.17), the "most significant" factor is $(\omega/n)^{|V(T)\cup A\cup B|}\omega^{-c}$, if notice

$$\frac{\binom{n-|V(T)\cup A\cup B|}{u-c}}{\binom{\omega-d+u}{u}} \omega^u n^{-(u-c)}$$

has 0th-order in ω, n. Other factors like $(|V(T)\cup A\cup B| + u - c + 8\tau^2)!/(8\tau^2)!$ are qualitatively smaller than ω, within our parameter regime.

4.1 Homogenization for Exact Clique

With Size Constraints (1.4) satisfied, any moment matrix can be reduced to its $\binom{[n]}{d/2}$-principal minor, which is slightly more convenient. The homogeneity trick is standard in SOS literature. Given a degree-d moment matrix $M^d_{\mathrm{SOS}}(G)$ satisfying (1.4), let $M(G)$ be its principal minor on $\binom{[n]}{d/2} \times \binom{[n]}{d/2}$.

Lemma 4.1. $M^d_{\mathrm{SOS}}(G)$ is PSD iff $M(G)$ is PSD.

Proof. The $\Rightarrow$ part is trivial. Now suppose M^d_{SOS} is not PSD, then

$$\exists a \in \mathbb{R}^{\binom{[n]}{\le d/2}} \quad a^\top M^d_{\mathrm{SOS}} a = -1. \tag{4.1}$$

With boolean constraints (define $\widetilde{\mathbb{E}}(x_i^2 p) := \widetilde{\mathbb{E}}(x_i p)$ for all i and polynomial p of degree $\le d-2$), this is equivalent to $\widetilde{\mathbb{E}}(g^2) = -1$ (4.2), where $g = a^\top x = \sum_{|S|\le d/2} a_S x_S$ is multi-linear. Now substitute every x_S ($|S| < d$) in g by the corresponding linear combination of $\{x_{S'} : |S'| = d\}$ from (3.13). This does not affect (4.2) since $\widetilde{\mathbb{E}}$ satisfies the equality constraints. We get $\widetilde{\mathbb{E}}(g_1^2) = -1$ (4.3) for some multi-linear, degree-$d/2$ homogeneous g_1. Translate (4.3) back (assume $g_1 = b^\top x$, $x = (x_S)_{|S|=d/2}$) to $b^\top M b = -1$, so M is not PSD. □

4.2 Concentration bound on polynomials

Lemma 4.2. Suppose $a < \log n$, and $p = \sum_{T:|V(T)|=a} c(T)\chi_T$, $c_T \in \mathbb{R}$, is a polynomial, and $C > 0$ satisfies $|c(T)| \le C$ for all T. Then w.p. $1 - n^{-10\log n}$ over G,

$$|p(G)| < C\, n^{a/2} 2^{a^2} n^{4\log\log n}. \tag{4.4}$$

Proof. Power-estimation. For all $k \in \mathbb{N}$ (think of $a < k = o(n/a)$),

$$p^{2k} = \sum_{T_1,\dots,T_{2k}:|V(T_i)|=a} c(T_1)\cdots c(T_{2k})\chi_{T_1}\cdots\chi_{T_{2k}}. \tag{4.5}$$

Take expectation. Each $\mathbb{E}[\chi_{T_1}\cdots\chi_{T_{2k}}(G)] \ne 0$ (equals 1) iff every edge appears even times in $T_1,\dots,T_{2k}$, which implies $|V(T_1\cup\cdots\cup T_{2k})| \le \frac{1}{2}\cdot 2ka = ka$. There are at most $\binom{n}{ka} < n^{ka}$ choices of such $V(T_1\cup\cdots\cup T_{2k})$. For each choice, at most $\binom{ka}{a}2^{\binom{a}{2}} < (ka)^a 2^{a^2/2}$ ways to choose each T_i. Therefore,

$$\mathbb{E}[p^{2k}] \le C^{2k} n^{ka}((ka)^a 2^{a^2/2})^{2k} =: N^{2k}, \quad N = Cn^{a/2}(ka)^a 2^{a^2/2}.$$

By Markov, $\Pr[p^{2k} > (2N)^{2k}] < 2^{-2k}$. Take $k := 10\log^2 n$; w.p. $> 1 - n^{-10\log n}$, $|p(G)| < 2N < Cn^{a/2}2^{a^2}n^{4\log\log n}$ for all large enough n. □

4.3 Norm concentration of pseudo-random matrices

Pseudo-random matrices refer to graphical matrices [24]: intuitively, such a matrix collects Fourier characters of all embeddings of a fixed "shape". Definitions4.3/4.5 are implicit in [24,25,16] and termed explicitly in [5].

Definition 4.3. A ribbon R is an ordered triple $(A, B; T)$ with A, B vertex-sets and T an edge-set. A, B are its left/right vertex-sets. Its size is $|V(R)| = |V(T)\cup A\cup B|$. By definition a ribbon as a graph has no isolated vertex outside $A \cup B$.

Definition 4.4. $R = (A, B; T)$ is left-generated if every vertex in $V(R)$ is either in B or can be reached by paths from A without touching B.[4] Right-generated is symmetric.

[4]The author always includes degenerate paths (one-point paths).

Definition 4.5. A ribbon $(A, B; T)$ is a shape if $A \cup B$ is totally ordered. Denote a shape by $U = (A, B; T)$, with $V(U) = A \cup B \cup V(T)$, size $|V(U)|$. When fixing underlying vertex-set $[n]$, a ribbon within $[n]$ can always be regarded as shapes with the induced vertex ordering. The author may interchangeably use R to denote shapes.

Definition 4.6. A real-valued function f on ribbons within $[n]$ is symmetric with respect to shapes if $f(R) = f(R')$ whenever R, R' have the same shape.

Definition 4.7 ([24]). Fix n and shape $U = (A, B; T)$. A map $\phi : V(U) \to [n]$ is proper if injective and respects the order on $A \cup B$. Its graphical matrix is the $2^{[n]} \times 2^{[n]}$ matrix

$$\forall I, J \subseteq [n], \quad M_U(I, J) = \sum_{T' : \exists \text{ proper } \phi,\ \phi(A)=I, \phi(B)=J, \phi(T)=T'} \chi_{T'}$$

(0 if no such ϕ exists). Here ϕ on T is the natural induced edge map.

Theorem 4.8 (Norm bounds on M_U [24,5]). For any shape $U = (A, B; T)$ of size $t < \log n$, w.p. $> 1 - n^{-10 \log n}$ over G,

$$\|M_U(G)\| \leq n^{(t-p)/2} 2^{O(t)} (\log n)^{O(t+p-2r)}, \tag{4.6}$$

where $r = |A \cap B|$, p is the maximum number of vertex-disjoint paths between (A, B) in U. This is tight up to polylog(n) factors for all M_U with these parameters ([24],Theorem38). Moreover, if $s = (|A| + |B|)/2$, then

$$\|M_U(G)\| \leq n^{(t-p)/2} 2^{O(t)} (\log n)^{O(t-s)}. \tag{4.7}$$

The author cites the standard trace-power norm theorem without reproducing its earlier proof. The "moreover" part follows since $t \geq |A \cup B| = 2s - r$, $p \leq s$, so $t + p - 2r \leq t + s - 2(2s - t) = 3(t - s)$.

Remark 4.9. Shapes $U = (A, A; T)$ give diagonal M_U, so estimating its norm is estimating the absolute values of its polynomial diagonals; Theorem4.8 generalizes Lemma4.2.

4.4 General notions on graphs

Definition 4.10 (Vertex-separator). Given graph H and $A, B \subseteq V(H)$, call $S \subseteq V(H)$ an (A, B)-vertex-separator if every path from A to B in H must pass through S.[5] Let $s_{A,B}(H) := \min\{|S| : S \text{ is an } (A, B)\text{-vertex-separator}\}$. A separator achieving this is a min-separator; $\text{mSep}_{A,B}(H)$ is the set of all min-separators. For ribbon $R = (A, B; T)$ use the graph on $V(T) \cup A \cup B$ with edges T and write $s_{A,B}(T)$, $\text{mSep}_{A,B}(T)$ or $\text{mSep}(R)$.

Menger's theorem. For finite graph H, $s_{A,B}(H)$ equals the maximum number of vertex-disjoint paths from A to B in H.

[5] Degenerate paths are included, as before.

Definition 4.11. The reduced size of ribbon $R = (A, B; T)$ is

$$e_{A,B}(T) := |V(T) \cup A \cup B| - s_{A,B}(T). \tag{4.8}$$

It is double the exponent of n in Theorem4.8's bound, hence controls the norm of the graphical matrix.

Required author Section 5.1: diagonalization of the expectation

Throughout Section5, M' denotes the $\binom{[n]}{d/2} \times \binom{[n]}{d/2}$ minor of the non-exact moment matrix:

$$M'(I, J) = \sum_{T:|V(T)\cup I\cup J|\le\tau} (\omega/n)^{|V(T)\cup I\cup J|}\chi_T, \quad \forall I, J \in \binom{[n]}{d/2}. \tag{5.1}$$

[6] The goal is to diagonalize M' approximately so that the difference matrix is negligible w.h.p. on plugging G.

Proposition 5.1. $\mathbb{E}[M'] = CC^\top$, where C is the $\binom{[n]}{d/2} \times \binom{[n]}{\le d/2}$ matrix

$$C = (\zeta^\top)_{d/2,\le d/2}\,\mathrm{diag}(\sqrt{t(|A|)})_{A\in\binom{[n]}{\le d/2}}, \tag{5.2}$$

and $t(r) = (1 - O(d\omega/n))(\omega/n)^{d-r}$ for $r = 0, \dots, d/2$.

Definition 5.2 (See e.g. [9]). Fix n, k. A Johnson scheme J is a $\binom{[n]}{k} \times \binom{[n]}{k}$ matrix satisfying $J(I, J) = J(I', J')$ whenever $|I \cap J| = |I' \cap J'|$. For fixed n, k, Johnson schemes are symmetric matrices and form a commutative $\mathbb{R}$-algebra, hence are simultaneously diagonalizable. Fix $k = d/2$. One basis is $D_0, \dots, D_{d/2}$ where

$$D_r(I, J) = \begin{cases} 1, & |I \cap J| = r, \\ 0, & \text{o.w.} \end{cases} \tag{5.3}$$

Another basis is $J_0, \dots, J_{d/2}$,

$$J_r(I, J) = \binom{|I \cap J|}{r}. \tag{5.4}$$

These are PSD since

$$J_r = \sum_{A\subseteq[n],|A|=r} u_A u_A^\top, \quad u_A \in \mathbb{R}^{\binom{[n]}{k}},\ u_A(B) = \mathbf{1}_{A\subseteq B}. \tag{5.5}$$

Clearly $J_{d/2} = \mathrm{Id}$.

[6] Strictly speaking PSDness of this minor is not sufficient since there is no homogeneity reduction in the non-exact case; it suffices to demonstrate the idea.

Fact 5.3 (See e.g. (4.29) in [9]). The Johnson schemes for $(n, d/2)$ have shared eigenspace-decomposition $\mathbb{R}^{\binom{[n]}{d/2}} = V_0 \oplus \cdots \oplus V_{d/2}$, and $J_r = \bigoplus_{i=0}^{d/2} \lambda_r(i)\Pi_i$, where Π_i is orthogonal projection to V_i w.r.t. Euclidean inner product, and

$$\lambda_r(i) = \binom{d/2-i}{r-i}\binom{n-d/2-i}{d/2-r}, \qquad 0 \le i \le d/2.$$

Lemma 5.4. $\mathbb{E}[M'] = \sum_{r=0}^{d/2} t(r) J_r$, with $t(r) = (1 - O(d\omega/n))(\omega/n)^{d-r}$.

Proof. By definition $\mathbb{E}[M'] = \sum_{r=0}^{d/2} (\omega/n)^{d-r} D_r$. Each D_r decomposes as

$$D_r = \sum_{r'=r}^{d/2} (-1)^{r'-r} \binom{r'}{r} J_{r'}. \tag{5.6}$$

Indeed the RHS at (I, J) equals

$$\begin{aligned}
&\sum_{r'=r}^{d/2} (-1)^{r'-r} \binom{r'}{r} \binom{|I \cap J|}{r'} \\
&= \sum_{r'=r}^{|I\cap J|} (-1)^{r'-r} \binom{|I \cap J|}{r} \binom{|I \cap J| - r}{r' - r} \\
&= \binom{|I \cap J|}{r} \mathbf{1}_{|I\cap J|=r} = \mathbf{1}_{|I\cap J|=r}.
\end{aligned}$$

So

$$\begin{aligned}
\mathbb{E}[M'] &= \sum_{r=0}^{d/2} (\omega/n)^{d-r} \left(\sum_{r'=r}^{d/2} (-1)^{r'-r} \binom{r'}{r} J_{r'} \right) \\
&= \sum_{r'=0}^{d/2} J_{r'} \left(\sum_{r=0}^{r'} (\omega/n)^{d-r} (-1)^{r'-r} \binom{r'}{r} \right) \\
&= \sum_{r'=0}^{d/2} J_{r'} (\omega/n)^{d-r'} (1 - \omega/n)^{r'}.
\end{aligned} \tag{5.7}$$

This proves the lemma. □

By Lemma5.4 and (5.5), letting $t(r') = (\omega/n)^{d-r'}(1-\omega/n)^{r'}$, we have

$$\begin{aligned}
\mathbb{E}[M'] &= \sum_{A : |A| \le d/2} t(|A|) u_A u_A^\top \\
&= (\zeta^\top)_{d/2, \le d/2} \operatorname{diag}(t(|A|)) \zeta_{\le d/2, d/2} = CC^\top,
\end{aligned}$$

since $(\zeta^\top)_{d/2,\le d/2}$ has columns $\{u_A : |A| \le d/2\}$. This proves Proposition5.1.

Fact 5.7 ([11], required separator notation from Section5.2). For ribbon $(I, J; T)$, all min-separators $\mathrm{mSep}_{I,J}(T)$ have a natural poset structure: $A_1 \leq A_2$ iff A_1 separates $(I, A_2; T)$, equivalently iff A_2 separates $(J, A_1; T)$. This poset is a lattice: each two min-separators have join and meet. In particular it has a unique minimum and maximum, denoted $S_l(I, J; T)$ and $S_r(I, J; T)$, the leftmost and rightmost min-separator. The rest of Section5.2 concerns the optional non-exact by-product; the author expressly permits skipping that section for the exact-clique proof.

Required author Section 5.3: recursive factorization

This formalizes and extends recursive factorization, giving notions extendable to matrix products (Definitions5.13/5.15), a simplification (Lemma5.25) and refinement (Proposition5.24) for later use.

Definition 5.9. Let L be the $\binom{[n]}{d/2} \times \binom{[n]}{\leq d/2}$ matrix

$$L(I, A) := \sum_{\substack{T' : |V(T') \cup I \cup A| \leq \tau,\ A = S_l(I, A; T') \\ T' \cap E(A) = \emptyset,\ (I, A; T') \text{ left-generated}}} (\omega/n)^{|V(T') \cup I \cup A| - |A|} \chi_{T'}. \tag{5.19}$$

Let Q_0 be the $\binom{[n]}{\leq d/2} \times \binom{[n]}{\leq d/2}$ matrix

$$Q_0(A, B) := \sum_{\substack{T_m : |T_m \cup A \cup B| \leq \tau \\ A, B \in \mathrm{mSep}_{A,B}(T_m)}} (\omega/n)^{|V(T_m) \cup A \cup B|} \chi_{T_m}. \tag{5.20}$$

Finally, $D := \mathrm{diag}((\omega/n)^{|A|/2})_{A \in \binom{[n]}{\leq d/2}}$ (5.21). The author calls $L(DQ_0)L^\top$ the first-approximate diagonalization of M'. Despite the name, $M' - L(DQ_0D)L^\top$ (5.22) is far from negligible. Recursive factorization will in the end give

$$M' = L[D(Q_0 - Q_1 + Q_2 \cdots \pm Q_{d/2})D]L^\top + E \tag{5.23}$$

for negligible error-matrix E.

Remark 5.10. Use of D is superficial; it keeps middle matrices Q_i better-positioned. LD corresponds to the "L" matrix of [5].

Definition 5.11 ([5],Definition6.5). The canonical decomposition of ribbon $R = (I, J; T)$ is the uniquely determined ribbon-triple

$$(R_l, R_m, R_r) = ((I, A; T_l), (A, B; T_m), (B, J; T_r)),$$

where $A = S_l(I, J; T)$, $B = S_r(I, J; T)$. $V(R_l)$ is A unioned with vertices reachable from I in T without touching A, and $T_l = T|_{V(R_l)} \setminus E(A)$. $V(R_r)$ is symmetric, reachable from J without touching B, with $T_r = T|_{V(R_r)} \setminus E(B)$. Finally $T_m = T \setminus (T' \sqcup T'')$. These are the left, middle and right ribbon of R.

Remark 5.12. The canonical decomposition satisfies

$$A = S_l(I, A; T_l), \quad B = S_r(B, J; T_r),$$
$$T_l \cap E(A) = \emptyset = T_r \cap E[A],$$
$$R_l \text{ left-generated}, \quad R_r \text{ right-generated},$$
$$A, B \in \mathrm{mSep}_{A,B}(T_m) \quad (\text{so } |A| = |B|).$$

These are the inner conditions. The outer intersection property is

$$V(R_l) \cap V(R_m) \subseteq A, \quad V(R_m) \cap V(R_r) \subseteq B, \quad V(R_l) \cap V(R_r) \subseteq A \cap B,$$

implying $e(R_l) + |V(R_m)| + e(R_r) = |V(R)|$ (5.24).

Definition 5.13 (Inner and outer canonicality). For a triple of ribbons $(R_l, R_m, R_r) = ((I, A; T_l), (A, B; T_m), (B, J; T_r))$ (T_l, T_m, T_r arbitrary subsets of an edge-set), their ribbon-sum is $(I, J; T)$ where $T = T_l \oplus T_m \oplus T_r$. It is inner-canonical if it satisfies the four inner conditions above (5.25). It is outer-canonical if

$$V(R_l) \cap V(R_m) \subseteq A, \quad V(R_m) \cap V(R_r) \subseteq B, \quad V(R_l) \cap V(R_r) \subseteq A \cap B. \tag{5.26}$$

It is canonical if both inner- and outer-canonical.

Proposition 5.14. Canonical triples are 1-1 correspondent to their ribbon-sum, via canonical decomposition.

Proof. This follows by an immediate check from the definition. □

Recall $\mathbb{R}[\{\chi_T\}]$ is the ring adding fresh variables χ_T into $\mathbb{R}$ for each $T \subseteq \binom{[n]}{2}$, with relations $\chi_{T'}\chi_{T''} = \chi_T$ whenever $T' \oplus T'' = T$.

Definition 5.15 (Approximate form). Suppose matrices X, Y have rows/columns indexed by subsets of $[n]$, entries in $\mathbb{R}[\{\chi_T\}]$, and every character viewed as a ribbon on (row,column) has size $\leq \tau$. Suppose $XYX^\top$ is defined. Every nonzero triple product, without collecting like terms, is

$$X(I, A; T_l)Y(A, B; T_m)X(J, B; T_r)\chi_{T_l}\chi_{T_m}\chi_{T_r} \tag{5.28}$$

in $XYX^\top$ (5.27), with the coefficient nonzero in $\mathbb{R}$. Identify it with the natural ribbon-triple; its resulting term is $X(I, A; T_l)Y(A, B; T_m)X(J, B; T_r)\chi_{T_l \oplus T_m \oplus T_r}$. Call it outer-canonical if that triple is outer-canonical; it exceeds degree if $|V(T) \cup I \cup J| > \tau$. The approximation form is

$$XYX^\top = (XYX^\top)_{\text{can}} + (XYX^\top)_{\text{non-can}} + E_{\text{deg}},$$
$$(XYX^\top)_{\text{can}} = XYX^\top - (XYX^\top)_{\text{non-can}} - E_{\text{deg}}. \tag{5.29}$$

Here the canonical part collects outer-canonical products not exceeding degree, the non-canonical part collects non-outer-canonical products, and E_{deg} all remaining terms.

Remark 5.16. Proposition5.14 explains Definition5.9 a posteriori: $M' = [L(DQ_0D)L^\top]_{\text{can}}$.

Definition 5.17 (First-approximate factorization of M').

$$M' = L(DQ_0D)L^\top - [L(DQ_0D)L^\top]_{\text{non-can}} - E_{1;\deg}. \tag{5.30}$$

$E_{1;\deg}$ is Definition5.15 applied to this product. $L(DQ_0D)L^\top$ is the first-approximate factorization. The high-degree error is negligible in norm (proved analogously in the exact case); matrices considered are supported on clique rows/columns of G. The main error is the non-canonical part. The key is that it too approximately factors through $L, L^\top$: there is Q_1 with

$$[L(DQ_0D)L^\top]_{\text{non-can}} = [L(DQ_1D)L^\top]_{\text{can}} + E'_{1;\text{negl}}. \tag{5.31}$$

Repeat for $[L(DQ_1D)L^\top]_{\text{non-can}}$ and so on.

Definition 5.18 ([5],Definition6.9). A generalized ribbon is a usual ribbon with a new isolated vertex-set. Write $R^* = (A, B; T^*)$ with $T^* = T \sqcup I$, T an edge-set, I disjoint from $V(T) \cup A \cup B$, denoted $I(R^*)$. $V(R^*) = V(T) \cup A \cup B \cup I$. Usual ribbons have $I = \emptyset$. $(A, B; T)$ is the unique largest ribbon in R.

Remark 5.19. $I(R^*)$ differs from the isolated set of the underlying graph since it excludes $A \cup B$.

Definition 5.20. A side-inner-canonical triple $((I, A; T_l), (A, B; T_m), (B, J; T_r))$ has left/right ribbons satisfying their inner-canonical conditions (first three of (5.25)); the middle is just a ribbon.

Definition 5.21 (Separating factorization; [5],Definition6.10). Given a side-inner-canonical triple (R_l, R_m, R_r) as above, put $T = T_l \oplus T_m \oplus T_r$. Let Z be the multiset of unexpected intersections, the vertices from $(R_l \cap R_m) - A$, $(R_m \cap R_r) - B$, $(R_l \cap R_r) - (A \cap B)$. Its intersection size is $z(R_l, R_m, R_r) = |Z|$. Then

$$|V(R_l) \cup V(R_m) \cup V(R_r)| = |V(R_l)| + |V(R_m)| + |V(R_r)| - |A| - |B| - z. \tag{5.32}$$

Define S'_l to be the leftmost min-separator of $(I, A \cup (Z \cap V(R_l)); T_l)$, and S'_r the rightmost min-separator of $(B \cup (Z \cap V(R_r)), J; T_r)$. They lie in $V(T) \cup I \cup J$. Define $R'_l = (I, S'_l; T'_l)$: $V(R'_l)$ is S'_l unioned with vertices in R_l reachable from I by paths in T_l without touching S'_l, and T'_l is $T_l \setminus E(S'_l)$ restricted to $V(R'_l)$. Define R'_r symmetrically. In particular $T'_l \cap T'_r = \emptyset$. The generalized middle ribbon is $R^*_m = (S'_l, S'_r; T^*_m)$ with

$$T^*_m = T \setminus (T'_l \sqcup T'_r) \sqcup I(R^*_m),$$

where $I(R^*_m)$ collects all remaining isolated vertices:

$$I(R^*_m) = V(R_l) \cup V(R_m) \cup V(R_r) - V(T) \cup I \cup J. \tag{5.33}$$

This separating factorization is denoted $(R_l, R_m, R_r) \to (R'_l, R^*_m, R'_r)$ (5.34).

Remark 5.22. (1) The resulting triple is side-inner-canonical and outer-canonical. Its pairwise intersections lie within $S'_l, S'_r, S'_l \cap S'_r$. Replacing R^*_m by its largest ribbon gives a canonical triple disjoint from $I(R^*_m)$. (2) $R'_l \subseteq R_l$, and S'_l separates $(V(R'_l), V(R_l) - V(R'_l))$ in R_l. The part of R_l to the right of S'_l is disjoint from R'_l and is in R^*_m; symmetrically for R_r. (3) Since S'_l separates (I, A) in R_l and A is its unique min-separator, there are $|A|$ vertex-disjoint paths from A to S'_l in R_l; symmetrically for R_r.

Required author factorization lemmas, Sections 5.3.2–5.3.3

Lemma 5.23 ([5],Lemmas6.14/7.14). Suppose $(R_l, R_m, R_r) \to (R'_l, R^*_m, R'_r)$. In Definition5.21's notation: (1) $|S'_l| + |S'_r| \geq |A| + |B| + 1$. (2) Put $s = (|A| + |B|)/2$, let p' be the maximum number of vertex-disjoint paths from S'_l to S'_r in R^*_m, and p from A to B in R_m. Then

$$2(s' - s) + (p - p') + |I(R^*_m)| \leq z(R_l, R_m, R_r).$$

Proof. (1) There must be an unexpected pairwise intersection. In any of the three cases breaking (5.26), there is $v \in Z$ in $V(R_l) - A$ or $V(R_r) - B$. WLOG the first. Then $S'_l \neq A$ since by left-generatedness v is reachable from I without passing A. Similarly, if $|S'_l| = |A|$ it is A, since A is the unique min-separator separating (I, A), impossible. Thus $S'_l > A$. (2) The author refers to the proof in the cited original paper. □

[7] The separating factorization cancels a term in $[L(DQ_0D)L^\top]_{\text{non-can}}$ by a canonical term using R'_l, R'_r and the largest ribbon of R^*_m in a new middle matrix DQ_1D. Other triples can have the same separating factorization, and entries of L are sums of different left ribbons. Simultaneous cancellation requires the following refined solvability condition, also needed in Lemma6.20.

Proposition 5.24 (Solvability condition, cf.[5],Claim6.12). Fix (I, J, S'_l, S'_r) and generalized ribbon R^*_m on (S'_l, S'_r). Let (R'_l, R'_r) be inner-canonical left/right ribbons on (I, S'_l), (S'_r, J). Let (R''_l, R''_r) be another such pair with $e(R'_l) = e(R''_l)$, $e(R'_r) = e(R''_r)$ (equivalently same sizes). For every fixed (i, j, z) there is a 1-1 matching between

$$\{(R_l, R_m, R_r) : (R_l, R_m, R_r) \to (R'_l, R^*_m, R'_r),\ (e(R_l), e(R_r), z(R_l, R_m, R_r)) = (i, j, z)\} \tag{5.35}$$

and the analogous set with (R''_l, R^*_m, R''_r) (5.36). The matching fixes each middle R_m.

Proof. Give a reversible map from (5.35) to (5.36). Take a triple in (5.35). By Remark5.22(2), the part of R_l right of S'_l is in R^*_m, hence disjoint from R'_l and R''_l; symmetrically for the right side. Map

$$(R_l, R_m, R_r) \mapsto (\phi(R_l), R_m, \phi(R_r)),$$

[7] In this setting R_m is a ribbon without isolated vertices. The credited prior lemma is a standard prerequisite, not a separately selected contribution.

where $\phi(R_l)$ replaces R_l' by R_l'' in R_l, and symmetrically on the right. R_m^*, hence R_m, is unchanged. Since R_l', R_l'' have the same size and the above parts are disjoint, this keeps $|V(R_l)|$. $R_l, \phi(R_l)$ have the same right distinguished set, the unique min-separator of both, so $e(R_l) = e(\phi(R_l))$; symmetrically on the right. Thus (i, j) is unchanged. z is unchanged since the changed part is disjoint from $Z(R_l, R_m, R_r)$. Reverse by switching the roles of the two ribbon pairs. □

Lemma 5.25 (One round of factorization). Let L be (5.19) and Q any $\binom{[n]}{\le d/2} \times \binom{[n]}{\le d/2}$ matrix with entries

$$Q(A,B) = \sum_{T_m : |V(T_m) \cup A \cup B| \le \tau} (\omega/n)^{|V(R_m)|} q(R_m) \chi_{T_m}, \tag{5.37}$$

where $R_m = (A, B; T_m)$ and q is symmetric w.r.t. shapes. Define Q' and E'_{negl} so that

$$(LQL^\top)_{\text{non-can}} = (LQ'L^\top)_{\text{can}} + E'_{\text{negl}}. \tag{5.38}$$

First set

$$Q'(A,B) = \sum_{T_m : |V(T_m) \cup A \cup B| \le \tau} (\omega/n)^{|V(R_m)|} q'(R_m) \chi_{T_m}. \tag{5.39}$$

Fix $R_m = (A, B; T_m)$ with $t = |V(R_m)| \le \tau$, $s = (|A| + |B|)/2$. For each generalized ribbon R_m^* whose largest ribbon is R_m and $|V(R_m^*)| \le \tau$, fix (R_l', R_r') such that (R_l', R_m^*, R_r') is the separating factorization of some triple with $|V(R_l')|, |V(R_r')| \le \tau$; exclude R_m^* if none. Let

$$q'(R_m) = \sum_{\substack{R_m^* : \text{generalized on } (A,B) \\ |V(R_m^*)| \le \tau,\ \text{largest ribbon } R_m}} (\omega/n)^{|I(R_m^*)|} q''(R_m^*),$$

$$q''(R_m^*) = \sum_{1 \le z \le d/2} \sum_{\substack{\mathcal{P} = (R_l, R, R_r) \text{: side-inner-canonical} \\ \mathcal{P} \to (R_l', R_m^*, R_r'),\ z(\mathcal{P}) = z}} (\omega/n)^z q(R). \tag{5.40}$$

By Proposition5.24 this is independent of the fixed pair and remains symmetric w.r.t. shapes. Define E'_{negl} by (5.38). Conclusions: (1) w.p. $> 1 - n^{-9 \log n}$,

$$\|E'_{\text{negl}}\| \le \max\{q(A,B;T)\}\, n^{-\epsilon\tau}.$$

(2) If C satisfies

$$\forall R_m \quad |q(R_m)| \le C(\omega/n^{1-\epsilon})^{s-p}, \tag{5.41}$$

where p is the maximum number of vertex-disjoint paths between A, B in R_m (equal to $s_{A,B}(T_m)$ by Menger), then

$$\forall R_m \quad |q'(R_m)| \le C(\omega/n^{1-\epsilon})^{s-p+1/3}.$$

Proof. Compare the canonical and non-canonical parts as (0), then prove (1),(2).

(0). For fixed (I, J), $[LQL^\top]_{\text{non-can}}(I, J)$ is

$$\sum_{\substack{(R_l, R_m, R_r) \text{:side-inner-canonical, non-outer-canonical} \\ \text{all three sizes} \le \tau}} (\omega/n)^{|V(R_l)| + |V(R_m)| + |V(R_r)| - |A| - |B|} q(R_m) \chi_{T_l \oplus T_m \oplus T_r}, \tag{5.42}$$

where (A, B) are the distinguished sets of R_m. Each triple has a unique separating factorization (R_l', R_m^*, R_r').

First case: $|V(R_m^*)| \le \tau$**.** The corresponding term in $(LQ'L^\top)_{\mathrm{can}}(I,J)$ is

$$(\omega/n)^{|V(R_l')|+|V(R_m')|+|V(R_r')|-|S_l'|-|S_r'|} \cdot (\omega/n)^{z+|I(R_m^*)|} q(R_m')\chi_{T_l'\oplus T_m^*\oplus T_r'}, \quad (5.43)$$

where R_m' is the largest ribbon of R_m^*, $\chi_{T_m^*}$ means the character from R_m', and $z \ge 1$ is the original intersection size. Recall $T_l' \oplus T_m^* \oplus T_r' = T_l \oplus T_m \oplus T_r$ and

$$\begin{aligned}|V(R_l)\cup V(R_m)\cup V(R_r)| &= |V(R_l')| + |V(R_m^*)| + |V(R_r')| - |S_l'| - |S_r'| \\ &= |V(R_l)| + |V(R_m)| + |V(R_r)| - |A| - |B| - z.\end{aligned}$$

Also $|V(R_m^*)| = |V(R_m')| + |I(R_m^*)|$. Together the coefficient in (5.43) equals the one in (5.42) from (R_l', R_m^*, R_r'). Conversely, by Q''s definition, (5.40) and Proposition5.24, each outer-canonical product corresponds uniquely to a side-inner-canonical triple of this case. Therefore the error collects precisely the next case.

Second case: $|V(R_m^*)| > \tau$**.** We have

$$E'_{\mathrm{negl}}(I,J) = \sum_{\substack{(R_l,R_m,R_r):\text{side-inner-canonical, non-outer-canonical}\\ \text{all three sizes}\le\tau,\ |V(R_m^*)|>\tau}} (\omega/n)^{|V(R_l)|+|V(R_m)|+|V(R_r)|-|A|-|B|} q(R_m)\chi_{T_l\oplus T_m\oplus T_r}. \quad (5.44)$$

The obvious left/right distinguished-set condition I, J is omitted.

(1). For such a triple, putting $T = T_l \oplus T_m \oplus T_r$, the exponent is

$$\begin{aligned}|V(R_l)| + |V(R_m)| + |V(R_r)| - |A| - |B| &= |V(R_l)\cup V(R_m)\cup V(R_r)| + z \\ &= |V(T)\cup I\cup J| + |I(R_m^*)| + z.\end{aligned}$$

Also $|I(R_m^*)| \le z + d/2$ from Lemma5.23.[8] Fix T and $a > \tau - |V(T)\cup I\cup J|$. To count triples contributing $(\omega/n)^{|V(T)\cup I\cup J|+a}\chi_T$ (temporarily ignore q), choose $I(R_m^*)$ of size $\le a/2 + d/4$, since $a = |I(R_m^*)| + z \ge 2|I(R_m^*)| - d/2$. Over fixed vertices the triple has $< 3^{3\tau}2^{3\binom{\tau}{2}}$ choices. With $B_0 = \max\{q(\cdot)\}$, its coefficient magnitude is less than

$$\begin{aligned}&B_0(\omega/n)^{|V(T)\cup I\cup J|+a} n^{(a+d)/2} 2^{2\tau^2} \\ &= B_0(\omega/n^{1-2\epsilon})^{|V(T)\cup I\cup J|} n^{-2\epsilon|V(T)\cup I\cup J|} (\omega/\sqrt{n})^a n^{d/2} 2^{2\tau^2} \\ &\le B_0(n^{-1/2})^{|V(T)\cup I\cup J|} n^{-2\epsilon(|V(T)\cup I\cup J|+a)} n^{d/2} 2^{2\tau^2} \\ &\le B_0(n^{-1/2})^{|V(T)\cup I\cup J|} n^{-1.5\epsilon\tau}.\end{aligned}$$

Use $\omega \le n^{1/2-4\epsilon}$, $|V(T)\cup I\cup J| + a > \tau$, $d < \epsilon\tau/10$, and $2^{2\tau} < n^{\epsilon/10}$. All appearing characters have $|V(T)| \le 3\tau$. By Lemma4.2, for fixed (I,J), w.p. $> 1 - n^{-10\log n}$,

$$|E'_{\mathrm{negl}}(I,J)| < \sum_{a=0}^{3\tau} B_0 n^{-a/2} n^{-1.5\epsilon\tau} n^{a/2} n^{4\log\log n} 2^{a^2} < n^{-1.4\epsilon\tau}.$$

[8] In fact $|I(R_m^*)| \le z$ can be shown, but is not needed.

Union-bound over $|\{(I,J)\}| < n^d$: w.p. $> 1 - n^{-9\log n}$, $\|E'_{\text{negl}}\| < n^d n^{-1.4\epsilon\tau} < n^{-\epsilon\tau}$.

(2). Fix R_m. By (5.40),

$$q'(R_m) = \sum_{z, R^*_m:\text{largest ribbon } R_m} (\omega/n)^{|I(R^*_m)|+z} \sum_{\substack{\mathcal{P}=(R_l,R,R_r):\text{side-inner-canonical} \\ \mathcal{P}\to(R'_l,R^*_m,R'_r),\ z(\mathcal{P})=z}} q(R).$$

For fixed R^*_m there are at most $8^{z\tau} < n^{\epsilon z}$ triples in the inner sum, since (R'_l, R'_r) is fixed: after choosing each vertex's occurrence in the ribbons and $A, B \subseteq R^*_m$, only edges appearing more than once remain to assign. Each such edge has an endpoint in the fixed multiset Z of size $\le z$. By Lemma5.23(2) and (5.41), the sum in absolute value is bounded by

$$\begin{aligned} &n^{\epsilon z}(\omega/n)^{z+|I(R^*_m)|}|q(R)| \\ &\le (\omega/n^{1-\epsilon})^{2(s'-s)+(p-p')+2|I(R^*_m)|} C(\omega/n^{1-\epsilon})^{s-p} \\ &\le C(\omega/n)^{2|I(R^*_m)|}(\omega/n^{1-\epsilon})^{s'-p'+1/2}. \end{aligned}$$

Here (s,p) are the parameters of R and (s',p') those of R_m; the last step uses $s'-s \ge 1/2$ from Lemma5.23(1). For fixed i_0 there are $< n^{i_0}$ choices of R^*_m with $|I(R^*_m)| = i_0$, and $1 \le z \le 3\tau$. Hence

$$\begin{aligned} |q'(R_m)| &\le 3\tau \sum_{i_0=0}^{d/2} C n^{i_0}(\omega/n)^{2i_0}(\omega/n^{1-\epsilon})^{s'-p'+1/2} \\ &\le C(\omega/n^{1-\epsilon})^{s'-p'+1/3}. \end{aligned}$$

□

Apply Lemma5.25 to the non-canonical term of (5.30), with $Q \leftarrow DQ_0D$: it becomes $[L(DQ_1D)L^\top]_{\text{can}} + E'_{1;\text{negl}}$. Repeat to obtain

$$\begin{aligned} M' = L[D(Q_0 - Q_1 + Q_2 - \cdots \pm Q_d)D]L^\top \\ - (E_{1;\text{deg}} - \cdots \pm E_{1+d;\text{deg}}) + (E'_{1;\text{negl}} + \cdots + E'_{d;\text{negl}}). \end{aligned} \tag{5.45}$$

Proposition 5.26 ([5],Claim6.15). $Q_{d+1} = 0$.

Proof. Induct that every ribbon $R_m = (A, B; T_m)$ in Q_k has $|A| + |B| \ge k$. The case $k = 0$ is trivial. Every ribbon in Q_{k+1} is the largest ribbon of a generalized middle in the separating factorization of a non-outer-canonical triple in $L(DQ_kD)L^\top$. The old middle has $|A| + |B| \ge k$, and Lemma5.23(1) gives $|A'| + |B'| \ge |A| + |B| + 1 \ge k + 1$. For $k = d + 1$ this is impossible with both distinguished sets in $\binom{[n]}{d/2}$. □

Remark 5.27. The final non-exact PSDness steps are omitted by the author, since similar arguments are given in the exact case, Section6. The selected target is the exact theorem; those complete Section6 steps follow below.

Author Section 6: PSDness of the exact pseudo-expectation

Henceforth M exclusively refers to the $d/2$-homogeneous minor of the moment matrix $\widetilde{M}$ in Definition3.13.

Theorem 6.1. W.p. $> 1 - n^{-5\log n}$,

$$M(G) \succeq n^{-d-1} \operatorname{diag}(\widetilde{\mathrm{Cl}}(G))_{\binom{[n]}{d/2}\times\binom{[n]}{d/2}}.$$

Corollary 6.2. W.p. $> 1 - n^{-5\log n}$, $\widetilde{\mathbb{E}}x_\emptyset > 0$.

Proof. By construction(3.13),

$$\widetilde{\mathbb{E}}x_\emptyset = \frac{\binom{\omega-d/2}{d-d/2}}{\binom{\omega-d}{d}\binom{d}{d/2}} \sum_{S:|S|=d/2} \widetilde{\mathbb{E}}x_S = \frac{\binom{\omega-d/2}{d-d/2}}{\binom{\omega-d}{d}\binom{d}{d/2}} \operatorname{Tr}(M),$$

positive w.h.p. by Theorem6.1. □

Theorem1.4 is a quick corollary: the pseudo-expectation from Definition3.13 has PSD moment matrix by Theorem6.1 and Lemma4.1; Default holds by Corollary6.2 and rescaling as discussed before Remark3.9; Clique and Size Constraints hold by Lemma3.8. The degree-d lower bound follows. The remainder proves Theorem6.1 by reducing it to Main Lemma6.9 and proving that lemma.

6.1 A Hadamard product and Euler transform

In $M(I,J)$ the parameter $u = |I \cap J|$ prevents termwise factorization like the non-exact case. We express M in a $\Sigma\Pi$-form where leaf matrices no longer depend on u, and factor those later.

6.1.1 Hadamard product

Put $u = |I \cap J|$, $a = |V(T) \cup I \cup J|$. From(3.17),

$$M(I,J;T) = \sum_{c=0}^{u} \left[\frac{\omega^{u-c}}{\binom{\omega-d+u}{u}} \mathcal{M}_c(u,a) \right], \tag{6.1}$$

where

$$\mathcal{M}_c(u,a) := \binom{a-(d-u)}{c} \binom{n-a}{u-c} n^{-(u-c)} \frac{(a+u-c+8\tau^2)!}{(8\tau^2)!} (\omega/n)^a.$$

Thus

$$M = \sum_{c=0}^{d/2} m_c \circ M_c, \tag{6.2}$$

with matrices, for $|I| = |J| = d/2$,

$$m_c(I,J) = \frac{\omega^{u-c}}{\binom{\omega-d+u}{u}}, \qquad u = |I \cap J|, \tag{6.3}$$

and

$$M_c(I,J) = \begin{cases} \sum_{T:|V(T)\cup I\cup J|\le\tau} \chi_T \mathcal{M}_c(|I\cap J|, |V(T)\cup I\cup J|), & |I\cap J|\ge c, \\ 0, & \text{o.w.} \end{cases} \tag{6.4}$$

Remark 6.3. m_c is supported on all (I,J), while $M_c(I,J)=0$ when $|I\cap J|<c$. The intuition is that M_c's are "close" for varying c, while m_c qualitatively decreases in c. This would let us concentrate on $c=0$. The next lemma proves the second half; Lemma6.9 states the first more precisely.

Lemma 6.4. For $c=0,\dots,d/2$,

$$m_c = \omega^{-c}\sum_{k=0}^{d/2} b_k J_k, \qquad b_k/k! \in [d/(2\omega), 1+2dk/\omega],$$

where J_k is (5.4). In particular,

$$m_0 = \omega m_1 = \cdots = \omega^{d/2} m_{d/2} \succ \frac{1}{\omega}\mathrm{Id}. \tag{6.5}$$

Proof. By definition,

$$m_c = \omega^{-c}\sum_{l=0}^{d/2} \frac{\omega^l}{\binom{\omega-d+l}{l}} D_l.$$

By basis-change(5.6), $m_c = \omega^{-c}\sum_{k=0}^{d/2} J_k b_k$ with

$$b_k = k!\left(\sum_{l=0}^{k}(-1)^{k-l}\underbrace{\frac{\omega}{\omega-(d-l)}\cdots\frac{\omega}{\omega-(d-1)}\frac{1}{(k-l)!}}_{=:f_k(l),\text{ which is } 1/k! \text{ if } l=0}\right).$$

For fixed k, $f_k(l)$ is increasing, so

$$\sum_{l=0}^{k}(-1)^{k-l}f_k(l) \ge f_k(k)-f_k(k-1) > \frac{d/2}{\omega}(1+(d/2)/\omega)^{k-1} \ge \frac{d}{2\omega}.$$

For $k=d/2$, $J_{d/2}=\mathrm{Id}$ gives(6.5). □

6.1.2 Euler transform

Fix c. For $(I,J;T)$ the coefficient $\mathcal{M}_c(u,a)$ is the partial function

$$\mathcal{M}_c(u,a) = \binom{a-(d-u)}{c}\binom{n-a}{u-c} n^{-(u-c)}\frac{(a+u-c+8\tau^2)!}{(8\tau^2)!}(\omega/n)^a. \tag{6.6}$$

Definition 6.5 (Extended $\mathcal{M}_c$). It is defined on $u \geq c$, $u + a \geq d + c$ and extended to $\mathbb{N}^2$ by

$$\binom{n-a}{u-c} = 0 \text{ if } u < c, \tag{6.7}$$

and the usual conventions

$$\binom{-m}{k} = (-1)^k \binom{m+k-1}{k} \quad (0 < m,\ 0 \leq k), \tag{6.8}$$

$$\binom{m}{k} = 0 \quad (0 \leq m < k). \tag{6.9}$$

In particular $\binom{m}{0} = 1$ for all $m \in \mathbb{Z}$, and if $0 \leq a-(d-u) < c$, then $\mathcal{M}_c(u, a) = 0$. Seek a decomposition

$$M_c = \sum_{R \in \binom{[n]}{\leq d/2}} M_c^R, \tag{6.10}$$

where M_c^R is supported on row/column indices containing R. Explicitly,

$$M_c^R(I, J) = \begin{cases} \sum_{T:|V(T)\cup I\cup J|\leq\tau} (\omega/n)^a Y_c(|R|, a)\chi_T, & R \subseteq I, J, \\ 0, & \text{o.w.} \end{cases} \tag{6.11}$$

where $a = |V(T) \cup I \cup J|$ and Y_c is to be chosen. Comparing each $(I, J; T)$, (6.10) is equivalent, for fixed c, a, to

$$\sum_{r=0}^{u} \binom{u}{r} Y_c(r, a)(\omega/n)^a = \mathcal{M}_c(u, a). \tag{6.12}$$

Thus take $Y_c(u, a)(\omega/n)^a$ to be the inverse Euler transform in u of the extended $\mathcal{M}_c$.

Fact 6.6. If sequences on $\mathbb{N}$ satisfy $x(m) = \sum_{l=0}^{m} \binom{m}{l} y(l)$, then x is the Euler transform of y, whose inverse is $y(m) = \sum_{l=0}^{m} (-1)^{m-l} \binom{m}{l} x(l)$.[9]

Definition 6.7 (Coefficients in M_c^R).

$$Y_c(r, a) = \begin{cases} \sum_{l=c}^{r} (-1)^{r-l} \binom{r}{l} \binom{a+l-d}{c} \binom{n-a}{l-c} n^{-(l-c)} \dfrac{(a+l-c+8\tau^2)!}{(8\tau^2)!}, & r \geq c, \\ 0, & \text{o.w.} \end{cases} \tag{6.13}$$

Lemma 6.8 (Hadamard-product decomposition).

$$M = \sum_{c=0}^{d/2} m_c \circ \left(\sum_{R} M_c^R \right) = \sum_{R \in \binom{[n]}{\leq d/2}} \underbrace{\left(\sum_{c=0}^{|R|} m_c \circ M_c^R \right)}_{=: M^R}. \tag{6.14–6.15}$$

Each m_c is Lemma6.4, and:

[9] This fact is an application of the ζ matrix and its inverse.

1. $M_c^R = 0$ if $|R| < c$.
2. If $R \not\subseteq I \cap J$, $M_c^R(I,J) = 0$.
3. If $|R| \geq c$ and $R \subseteq I \cap J$, then $M_c^R(I,J) = \sum_{T:|V(T)\cup I\cup J|\leq\tau} M_c^R(I,J;T)\chi_T$, with
$$\begin{aligned} M_c^R(I,J;T) &= (\omega/n)^a \sum_{l=c}^{|R|} (-1)^{|R|-l} \binom{|R|}{l} \binom{a+l-d}{c} \\ &\quad \cdot \binom{n-a}{l-c} n^{-(l-c)} \frac{(a+l-c+8\tau^2)!}{(8\tau^2)!} \\ &= (\omega/n)^a Y_c(|R|,a). \end{aligned} \tag{6.16}$$
4. For $0 \leq c \leq r \leq d/2$, $0 \leq a \leq \tau$, $|Y_c(r,a)| < \tau^{5\tau}$.

Proof. (1),(2),(3) are definition. To check $M_c = \sum_R M_c^R$, fix $(I,J;T)$ with $|I| = |J| = d/2$, $a = |V(T) \cup I \cup J| \leq \tau$, and $u = |I \cap J|$. Note $a - (d-u) \geq 0$ and

$$\sum_R M_c^R(I,J;T) = \sum_{R\subseteq I\cap J} M_c^R(I,J;T) = (\omega/n)^a \sum_{r=0}^{|I\cap J|} \binom{|I\cap J|}{r} Y_c(r,a).$$

By Euler transform and(6.12) this equals extended $\mathcal{M}_c(u,a)$. It only remains to see $\mathcal{M}_c(u,a) = 0$ if $u < c$ or $a - (d-u) < c$, by(6.7),(6.9). For(4),

$$\begin{aligned} |Y_c(u,a)| &= \left| \sum_{l=c}^{r} (-1)^{r-l} \binom{r}{l} \binom{a+l-d}{c} \left[\binom{n-a}{l-c} n^{-(l-c)} \right] \frac{(a+l-c+8\tau^2)!}{(8\tau^2)!} \right| \\ &< r\, 2^r (2\tau)^r \cdot 1 \cdot (9\tau^2)^{2\tau} < \tau^{5\tau}, \end{aligned}$$

using $r \leq d/2 \ll \tau$ in our regime. □

Lemma 6.9 (Main Lemma). In(6.15), w.p. $> 1 - n^{-5\log n}$, for every $R \in \binom{[n]}{\leq d/2}$, with $\mathcal{P}^R = \{I \in \binom{[n]}{d/2} : R \subseteq I\}$,

$$M_0^R \succeq n^{-d} \operatorname{diag}(\widetilde{\mathrm{Cl}})_{\mathcal{P}^R \times \mathcal{P}^R}, \tag{6.17}$$

and

$$\pm\omega^{-c} M_c^R \preceq n^{-c/6} M_0^R \quad \forall 0 < c \leq |R|. \tag{6.18}$$

Corollary 6.10 (Theorem6.1). W.p. $> 1 - n^{-5\log n}$, $M(G) \succeq n^{-d-1} \operatorname{diag}(\widetilde{\mathrm{Cl}}(G))$ on $\binom{[n]}{d/2} \times \binom{[n]}{d/2}$.

Proof. Suppose Lemma6.9 holds. For fixed R, $M^R = \sum_{c=0}^{|R|} m_c \circ M_c^R$. Hadamard product with a PSD matrix preserves PSDness (Schur product theorem), so

$$\begin{aligned} \sum_{c=1}^{|R|} m_c \circ M_c^R &\preceq \sum_{c=1}^{|R|} m_c \circ (\omega^c n^{-c/6} M_0^R) \\ &= \left(\sum_{c=1}^{|R|} n^{-c/6} m_0 \right) \circ M_0^R \\ &\preceq n^{-1/6} m_0 \circ M_0^R. \end{aligned}$$

Similarly the sum is $\succeq -n^{-1/6} m_0 \circ M_c^R$. Thus

$$M^R \succeq (1 - n^{-1/6}) m_0 \circ M_0^R \succeq n^{-d-1} \operatorname{diag}(\widetilde{\mathrm{Cl}})_{\mathcal{P}^R \times \mathcal{P}^R}$$

by Lemmas6.4/6.9(2). Apply(6.15):

$$M = M^{\emptyset} + \sum_{\emptyset \neq R \in \binom{[n]}{\leq d/2}} M^R \succeq M^{\emptyset} \succeq n^{-d-1} \operatorname{diag}(\widetilde{\mathrm{Cl}})_{\binom{[n]}{d/2} \times \binom{[n]}{d/2}}. \tag{6.19}$$

□

Author Section 6.2: first-approximate factorization of M_c^R

Use Section5.3's terminology and extend the recursive approximate factorization.

Definition 6.11. Fix $R \in \binom{[n]}{\leq d/2}$. For $i = 0, \ldots, \tau$, define the left-i-factor $L_{R,i}$, of dimension $\binom{[n]}{d/2} \times \binom{[n]}{\leq d/2}$, by

$$L_{R,i}(I, A) = \begin{cases} 0, & R \not\subseteq I \cap A, \\ \displaystyle\sum_{\substack{T : |V(T) \cup I \cup A| \leq \tau,\ A = S_l(I,A;T) \\ T \cap E(A) = \emptyset,\ (I,A;T) \text{ left-generated} \\ e_{I,A}(T) = i}} (\omega/n)^i \chi_T, & \text{o.w.} \end{cases} \tag{6.20}$$

$(L_{R,j})^\top$ is the right-j-factor. $\widetilde{L}_R = (L_{R,0}, \ldots, L_{R,\tau})$ is the left factor and $(\widetilde{L}_R)^\top$ the right factor. These do not depend on c.

Definition 6.12. Let D^τ be the constant diagonal matrix

$$\operatorname{diag}((\omega/n)^{|A|/2})_{A : |A| \leq d/2} \otimes \mathrm{Id}_{\{0,\ldots,\tau\} \times \{0,\ldots,\tau\}},$$

of dimension $(\binom{[n]}{\leq d/2} \times (\tau + 1)) \times (\binom{[n]}{\leq d/2} \times (\tau + 1))$.

Definition 6.13 (Goal factorization). Find a middle matrix Q_c^R of that same dimension such that approximately

$$M_c^R \approx \widetilde{L}_R [D^\tau Q_c^R D^\tau] (\widetilde{L}_R)^\top. \tag{6.21}$$

Remark 6.14. Unlike Section5.3, further distinguish a parameter pair in $\{0, \ldots, \tau\}^2$: in(6.13), and any exact pseudo-expectation generated by Section3.2, $a = |V(T) \cup I \cup J|$ appears nestedly. Previously $(\omega/n)^a = (\omega/n)^{e(R_l) + |V(R_m)| + e(R_r)}$ naturally factors left/middle/right. Here terms like $\binom{a+l-d}{c}\binom{n-a}{l-c}$ are not log-additive in a. The chosen generating function makes $\mathbb{E}[Q_{0,0}^R]$ positive in this harder situation, as Proposition6.28/Corollary6.30 will show. Separate the main factor $(\omega/n)^a$ as before, and leave $Y_c(r, a)$ to the middle matrix indexed by (e_l, e_r).

Definition 6.15 (First-approximate factorization). $Q^R_{c,0}$ is a $(\tau+1)\times(\tau+1)$ block matrix, each block of dimension $\binom{[n]}{\le d/2}\times\binom{[n]}{\le d/2}$, zero outside the principal minor $\mathcal{S}^R\times\mathcal{S}^R$ with

$$\mathcal{S}^R := \{(A,i)\in\binom{[n]}{\le d/2}\times\{0,\dots,\tau\} : A\supseteq R,\ |A|+i\ge d/2\}. \tag{6.22}$$

Inside this minor,

$$Q^R_{c,0}((A,i),(B,j)) = \sum_{\substack{T_m:|V(T_m)\cup A\cup B|\le\tau\\ A,B\in\mathrm{mSep}_{A,B}(T_m)}} (\omega/n)^{|V(T_m)\cup A\cup B|-(|A|+|B|)/2} \cdot Y_c(|R|,|V(T_m)\cup A\cup B|+(i+j))\chi_{T_m}. \tag{6.23}$$

Define $\widetilde{L}_R[D^\tau Q^R_{c,0}D^\tau](\widetilde{L}_R)^\top$ as the first approximate factorization of M^R_c.

Remark 6.16. (1) $\mathcal{S}^R$ is independent of c; $|A|+i\ge d/2$ is natural since i is intended as $|V(T')\setminus A|\ge|I|-|A|$ for a left ribbon. Otherwise that column in $\widetilde{L}_R$ is zero. Similarly for j. (2) $Q^R_{c,0}$ is supported where $|A|=|B|$. (3) For an outer-canonical product in the above matrix product, Remark5.12 gives $|V(T_m)\cup A\cup B|+(i+j)=|V(T)\cup I\cup J|$ for the canonical decomposition with $e(R_l)=i$, $e(R_r)=j$.

Lemma 6.17. Fix $R, c\le|R|$. For each $(I,J;T)$ with $|V(T)\cup I\cup J|\le\tau$ and $R\subseteq I\cap J$, there is exactly one outer-canonical product in

$$\widetilde{L}_R[D^\tau Q^R_{c,0}D^\tau](\widetilde{L}_R)^\top \tag{6.24}$$

corresponding to its canonical decomposition, giving the term $M^R_c(I,J;T)\chi_T$.

Proof. Every triple in(6.24) is inner-canonical by the definitions of $\widetilde{L}_R, Q^R_{c,0}$, hence outer-canonical triples 1-1 correspond to $(I,J;T)$ by canonical decomposition. Fix such a decomposition. $(I,A;T')$ appears once in $\widetilde{L}_R(I,A)$ in block L_{R,e_l}, $e_l=e_{I,A}(T')$; symmetrically for $(J,B;T'')$, $e_r=e_{J,B}(T'')$. Its unique outer-canonical product coefficient is

$$L_{R,e_l}(I,A;T')(\omega/n)^{|A|/2}Q^R_{c,0}(A,B;T_m)(\omega/n)^{|B|/2}L_{R,e_r}(J,B;T''). \tag{6.25}$$

By(6.20),(6.23), if $a=|V(T)\cup I\cup J|\le\tau$, this equals $(\omega/n)^a Y_c(|R|,a) = M^R_c(I,J;T)$ by(6.13),(6.16), since canonicality gives $a=e_l+|V(T_m)\cup A\cup B|+e_r$. □

Definition 6.18 (First error-matrices). Let $E_{c,1;\mathrm{negl}}$ sum all outer-canonical products in(6.24) exceeding degree, namely $|V(T)\cup I\cup J|>\tau$. Let the non-canonical part sum all non-outer-canonical products. Lemma6.17 says $M^R_c=[\widetilde{L}_R(D^\tau Q^R_{c,0}D^\tau)(\widetilde{L}_R)^\top]_{\mathrm{can}}$, equivalently

$$\begin{aligned} M^R_c &= \widetilde{L}_R(D^\tau Q^R_{c,0}D^\tau)(\widetilde{L}_R)^\top \\ &\quad - [\widetilde{L}_R(D^\tau Q^R_{c,0}D^\tau)(\widetilde{L}_R)^\top]_{\text{non-can}} - E^R_{c,1;\deg}. \end{aligned} \tag{6.26}$$

The crucial fact is that the main error approximately factorizes through the same left/right factors; the next subsection extends recursive factorization to do this.

Author Section 6.3: exact recursive factorization statement

Lemma 6.19 (Recursive approximate factorization; exact case). For any fixed $R \in \binom{[n]}{\le d/2}$ and $0 \le c \le |R|$,

$$M_c^R = \widetilde{L}_R[D^\tau(Q_{c,0}^R - Q_{c,1}^R + \cdots \pm Q_{c,d}^R)D^\tau](\widetilde{L}_R)^\top + E_c^R, \tag{6.27}$$

where:

1. All $Q_{c,k}^R$ are supported on $\mathcal{S}^R \times \mathcal{S}^R$, with $\mathcal{S}^R$ from(6.22).
2. $Q_{c,0}^R$ is Definition6.15.
3. For $1 < k \le d/2$, within that minor the (i,j) block is

$$Q_{c,k}^R((A,i),(B,j)) = \sum_{T_m : |V(T_m)\cup A\cup B| \le \tau} q_{c,k}^R(R_m,i,j)\chi_{T_m}, \tag{6.28}$$

   $R_m = (A,B;T_m)$, the coefficients symmetric w.r.t. shapes, and

$$\forall (i,j) \quad |q_{c,k}^R(R_m,i,j)| \le \tau^{5\tau}(\omega/n^{1-\epsilon})^{s-p+k/3}, \tag{6.29}$$

   where $s = (|A|+|B|)/2$ and p is the maximum number of vertex-disjoint paths from A to B in R_m.
4. For any G, $E_c^R(G)$ is supported on clique rows/columns containing R, and w.p. $> 1 - n^{-9\log n}$,

$$\|E_c^R\| < n^{-\epsilon\tau/2}. \tag{6.30}$$

The proof proceeds via one-round Lemma6.20, followed by its iteration; the complete author proof is continued in the next fragment.

Complete author exact recursive-factorization proof

Again fix $R \in \binom{[n]}{\le d/2}$, $c \le |R|$, and write $n_1 := \binom{[n]}{d/2} \times (\tau+1)$ for convenience.

Lemma 6.20 (One round of factorization; exact case). Let $\widetilde{L}_R$ be Definition6.11 and Q^R any $n_1 \times n_1$ matrix supported on $\mathcal{S}^R \times \mathcal{S}^R$, with

$$Q^R((A,i),(B,j)) = \sum_{T_m : |V(T_m)\cup A\cup B| \le \tau} (\omega/n)^{|V(R_m)|} q(R_m,i,j)\chi_{T_m}, \tag{6.31}$$

where $R_m = (A,B;T_m)$, $q(\cdot,i,j)$ symmetric w.r.t. shapes for fixed (i,j). Define Q' and E'_{negl} so that

$$[\widetilde{L}_R Q (\widetilde{L}_R)^\top]_{\text{non-can}} = [\widetilde{L}_R Q' (\widetilde{L}_R)^\top]_{\text{can}} + E'_{\mathrm{negl}}. \tag{6.32}$$

Let Q' be supported on $\mathcal{S}^R \times \mathcal{S}^R$ and

$$Q'((A,i),(B,j)) = \sum_{T_m : |V(T_m)\cup A\cup B| \le \tau} (\omega/n)^{|V(R_m)|} q'(R_m,i,j)\chi_{T_m}. \tag{6.33}$$

For fixed $R_m = (A, B; T_m)$, (i, j), put $t = |V(R_m)| \le \tau$, $s = (|A| + |B|)/2$. For every generalized R_m^* with largest ribbon R_m and size $\le \tau$, fix a pair (R_l', R_r') such that (R_l', R_m^*, R_r') is a separating factorization of some triple, $|V(R_l')|, |V(R_r')| \le \tau$ and

$$(e(R_l'), e(R_r')) = (i, j). \tag{6.34}$$

Exclude R_m^* if no choice exists. Set

$$\begin{aligned} q'(R_m, i, j) &= \sum_{\substack{R_m^*:\text{generalized on } (A,B) \\ |V(R_m^*)|\le\tau,\ \text{largest ribbon } R_m}} (\omega/n)^{|I(R_m^*)|} q''(R_m^*, i, j), \\ q''(R_m^*, i, j) &= \sum_{z, i_1, j_1:\ 1\le z\le d/2} \sum_{\substack{\mathcal{P}=(R_l, R, R_r):\text{side-inner-canonical} \\ \mathcal{P}\to(R_l', R_m^*, R_r'),\ z(\mathcal{P})=z \\ e(R_l)=i_1,\ e(R_r)=j_1}} (\omega/n)^z q(R, i_1, j_1). \end{aligned} \tag{6.35}$$

The full Proposition5.24 makes this independent of the fixed pair, so q' is symmetric w.r.t. shapes. Define the error by(6.32). Conclusions: (1) w.p. $> 1 - n^{-9\log n}$, $\|E'_{\text{negl}}\| \le \max\{q(\cdot)\} n^{-\epsilon\tau}$. (2) If

$$\forall R_m, i, j \quad |q(R_m, i, j)| \le C(\omega/n^{1-\epsilon})^{s-p}, \tag{6.36}$$

where p is maximum vertex-disjoint A–B path count, then $|q'(R_m)| \le C(\omega/n^{1-\epsilon})^{s-p+1/3}$ for all R_m, i, j.

Proof of Lemma 6.20. The proof is almost the same as Lemma5.25; the author gives all differences as follows. The support condition does not affect anything, since $\widetilde{L}_R$ is automatically zero on columns/rows not in $\mathcal{S}^R$. Expand the canonical product with Q' to compare termwise to the non-canonical product with Q, by Proposition5.24. When (i, j) and R_m^* are fixed, the sizes of any pair satisfying(6.34) are fixed, so that proposition applies. The comparison of orders in ω/n is exactly step(0) of Lemma5.25. Thus the error collects all non-canonical terms whose separating middle exceeds size τ:

$$\begin{aligned} E'_{\text{negl}}(I, J) = \sum_{i,j} \sum_{\substack{(R_l, R_m, R_r):\text{side-inner-canonical, non-outer-canonical} \\ \text{all three sizes}\le\tau,\ |V(R_m^*)|>\tau \\ (e(R_l), e(R_r))=(i,j)}} & (\omega/n)^{|V(R_l)|+|V(R_m)|+|V(R_r)|-|A|-|B|} \\ & \cdot q(R_m, i, j)\chi_T, \\ & T = T_l \oplus T_m \oplus T_r. \end{aligned} \tag{6.37}$$

Left/right distinguished sets I, J are implicit. Numerical conclusions(1),(2) follow from the estimates after(5.44). For(1), that estimate is loose enough that even an extra $(1+\tau)^2$ factor from union-bound over blocks remains below $n^{-\epsilon\tau}$. □

Proof of Lemma 6.19. Apply Lemma6.20 to $[\widetilde{L}_R(D^\tau Q^R_{c,i} D^\tau)(\widetilde{L}_R)^\top]_{\text{non-can}}$ for $i = 0$, obtaining $Q^R_{c,1}, E'_{1;\text{negl}}$ (hide R, c on this negligible matrix). Repeat for $i = 1$ obtaining $E_{c,1;\deg}, Q^R_{c,2}, E'_{2;\text{negl}}$. Continuing gives

$$\begin{aligned} M^R_c = \widetilde{L}_R[D^\tau(Q^R_{c,0} - Q^R_{c,1} + \cdots \pm Q^R_{c,d})D^\tau](\widetilde{L}_R)^\top \\ - (E^R_{c,1;\deg} - E^R_{c,2;\deg} + \cdots \pm E^R_{c,d;\deg}) \\ + (E'_{1;\text{negl}} + \cdots + E'_{d;\text{negl}}). \end{aligned} \tag{6.38}$$

Here $Q^R_{c,d+1} = 0$ by the same Proposition5.26.

1. All $Q^R_{c,k}$ have the required support by the definition of each round.
2. By definition.
3. Each coefficient family remains symmetric w.r.t. shapes by Lemma6.20. From(6.23) and Lemma6.8(4), $|q^R_{c,0}(R_m)| = |Y_c(|R|, |R_m|)| \le \tau^{5\tau}$. Every ribbon appearing in $Q^R_{c,0}$ has $|A| = |B|$ vertex-disjoint A–B paths, so $s = p$. Equivalently,
$$|q^R_{c,0}(R_m)| \le (\omega/n^{1-\epsilon})^{s-p}\tau^{5\tau}. \tag{6.39}$$
Use Lemma6.20(2): its q corresponds to $q^R_{c,k}$, because its matrix Q is $D^\tau Q^R_{c,k} D^\tau$, whose $(\omega/n)^{|V(R_m)|}q$ equals $(\omega/n)^{|V(R_m)|-s}(\omega/n)^s q^R_{c,k}$. Thus recursively
$$|q^R_{c,k}(R_m, i, j)| \le \tau^{5\tau}(\omega/n^{1-\epsilon})^{s-p+k/3}.$$
4. For every G, both M^R_c and its factored expression are supported on clique rows/columns containing R, so their difference E^R_c is too. From(6.38) it is the sum of alternating degree errors and negligible errors. By Lemma6.20(2), inductively all $|q^R_{c,k}| < \tau^{5\tau}$. By Lemma6.20(1), each $E'_{k;\text{negl}}$ satisfies w.p. $> 1 - n^{-9\log n}$,
$$\|E'_{k;\text{negl}}\| < \tau^{5\tau} n^{-\epsilon\tau} < n^{-0.9\epsilon\tau}.$$

For the degree error, Definition5.15 says $E^R_{c,k;\deg}(I, J)$ sums all outer-canonical products exceeding degree in $\widetilde{L}_R(D^\tau Q^R_{c,i-1} D^\tau)(\widetilde{L}_R)^\top$. Thus

$$E^R_{c,k;\deg}(I, J) = \sum_{\substack{(R_l, R_m, R_r):\text{semi-inner-canonical, outer-canonical} \\ |V(T)\cup I\cup J|>\tau}} (\omega/n)^{|V(T)\cup I\cup J|} \cdot q^R_{c,k-1}(R_m, e(R_l), e(R_r))\chi_T.$$

As usual $T = T_l \oplus T_m \oplus T_r$, side sets are I, J, and s is the average size of the middle distinguished sets. Outer/semi-inner canonicality gives $|V(T) \cup I \cup J| = e_l + e_r + |V(R_m)|$. Each fixed $(I, J; T)$ can come from at most $3^{3\tau}$ triples, since their vertex-set union is exactly $V(T) \cup I \cup J$. Since $3\tau \ge |V(T) \cup I \cup J| > \tau$ and w.h.p. $|q^R_{c,k-1}| < \tau^{5\tau}$, Lemma4.2 gives w.p. $> 1 - n^{-10\log n}$,

$$|E^R_{c,k;\deg}(I, J)| < \tau^{6\tau} \sum_{c=0}^{3\tau} (\omega/n)^{\max\{\tau,c\}} n^{c/2} 2^{c^2} n^{4\log\log n} < n^{-2\epsilon\tau}.$$

Union-bound over (I, J) gives $\|E^R_{c,k;\deg}\| < n^{-d/4} n^{-2\epsilon\tau} < n^{-\epsilon\tau}$ w.p. $> 1 - n^{-9.5\log n}$. Sum the two errors and union-bound over k: w.p. $> 1 - n^{-9\log n}$, $\|E^R_c\| < n^{-\epsilon\tau/2}$. □

Author Sections 6.4–6.5: positivity of middle matrices

Write $Q_c^R = Q_{c,0}^R - Q_{c,1}^R + \cdots \pm Q_{c,d}^R$ in Lemma6.19's approximate decomposition. The following is the key lemma, proved through the next subsections.

Lemma 6.21. W.p. $> 1 - n^{-8\log n}$: (1) for all $R \in \binom{[n]}{\le d/2}$,

$$Q_{0,0}^R - Q_{0,1}^R + \cdots \pm Q_{0,d/2}^R \succeq \tau^{-7\tau}\,\mathrm{diag}(\widetilde{\mathrm{Cl}})_{\mathcal{S}^R\times\mathcal{S}^R};$$

(2) for all R, $0 < c \le |R|$,

$$\pm\omega^{-c}(Q_{c,0}^R - Q_{c,1}^R + \cdots \pm Q_{c,d/2}^R) \preceq n^{-c/4}\,\mathrm{diag}(\widetilde{\mathrm{Cl}})_{\mathcal{S}^R\times\mathcal{S}^R}.$$

The proof has three ingredients: Corollary6.36 shows positivity of $Q_{0,0}^R$ via a structural/pseudorandom decomposition; Lemmas6.37/6.38 bound the rest. The last real technical challenge is $\mathbb{E}[Q_{0,0}^R]$, ultimately the choice of F in Definition3.11.

Definition 6.22. The root diagonal-clique matrix on $\binom{[n]}{\le d/2}$ is

$$D_{\mathrm{Cl}}(A,B) = \begin{cases} 0, & A \ne B, \\ 2^{-\binom{|A|}{2}/2}\widetilde{\mathrm{Cl}}_A = 2^{-\binom{|A|}{2}/2}\sum_{T\subseteq E[A]}\chi_T, & A = B. \end{cases} \tag{6.40}$$

Thus $D_{\mathrm{Cl}}^2(A,A) = \widetilde{\mathrm{Cl}}(A)$. Define $D_{\mathrm{Cl}}^\tau := D_{\mathrm{Cl}} \otimes \mathrm{Id}_{\{0,\dots,\tau\}\times\{0,\dots,\tau\}}$ (6.41), also diagonal.

Definition 6.23. The structural-pseudorandom decomposition is

$$Q_{0,0}^R = D_{\mathrm{Cl}}^\tau \mathbb{E}[Q_{0,0}^R] D_{\mathrm{Cl}}^\tau + \left(Q_{0,0}^R - D_{\mathrm{Cl}}^\tau \mathbb{E}[Q_{0,0}^R] D_{\mathrm{Cl}}^\tau\right). \tag{6.42}$$

The first summand is structural and the second pseudorandom.

Proposition 6.24 (Expression of $\mathbb{E}[Q_{c,0}^R]$). Fix R, $0 \le c \le |R|$, $r = |R|$.

1. The expectation is supported on blockwise partial diagonals $\{((A,i),(A,j)) \in \mathcal{S}^R \times \mathcal{S}^R\}$, requiring $R \subseteq A$ and $|A| + \min\{i,j\} \ge d/2$.
2. On these positions,

$$\mathbb{E}[Q_{c,0}^R]((A,i),(A,j)) = \sum_{l=c}^{r}(-1)^{r-l}\frac{\binom{r}{l}}{(l-c)!}\binom{|A|+i+j+l-d}{c} \cdot \frac{(|A|+8\tau^2+(l-c)+(i+j))!}{(8\tau^2)!} + O(\tau^{1.5\tau}/n). \tag{6.43}$$

For $c = 0$ this is

$$\sum_{l=0}^{r}(-1)^{r-l}\frac{\binom{r}{l}}{l!}\frac{(|A|+8\tau^2+l+(i+j))!}{(8\tau^2)!} + O(\tau^{1.5\tau}/n). \tag{6.44}$$

3. If $1_{A,A}$ is the matrix with one 1 at (A,A), then

$$\mathbb{E}[Q^R_{0,0}] = \sum_{\substack{A\in\binom{[n]}{\le d/2}\\ A\supseteq R}} 1_{A,A}\otimes\left[\sum_{l=0}^{r}(-1)^{r-l}\frac{\binom{r}{l}}{l!}P_{|A|+l}+E^R_A\right]. \tag{6.45}$$

$P_{|A|+l}$ and E^R_A are $(\tau+1)\times(\tau+1)$ matrices supported on $\{i : d/2-|A|\le i\le\tau\}^2$,

$$\|E^R_A\| < \tau^{2\tau}/n, \tag{6.46}$$

and

$$P_{|A|+l}(i,j) = \frac{(|A|+l+8\tau^2+(i+j))!}{(8\tau^2)!}. \tag{6.47}$$

Proof. (1) Constant terms in(6.23) have $T_m=\emptyset$, nonzero only when $A=B$. (2) For those positions the expectation is $Y_c(r,a)$, $a=|A|+i+j$, namely

$$\sum_{l=c}^{r}(-1)^{r-l}\binom{r}{l}\binom{a+l-d}{c}\binom{n-a}{l-c}n^{-(l-c)}\frac{(a+l-c+8\tau^2)!}{(8\tau^2)!}. \tag{6.48}$$

Use

$$\binom{n-a}{l-c}n^{-(l-c)} = \frac{1}{(l-c)!}\frac{(n-a)\cdots(n-a-(l-c)+1)}{n^{l-c}} = \frac{1}{(l-c)!}(1-O(d^2/n)),$$

and the bound

$$\binom{r}{l}\binom{a+l-d}{c}\binom{n-a}{l-c}n^{-(l-c)}\frac{(a+l-c+8\tau^2)!}{(8\tau^2)!} < (4d)^d(9\tau^2)^d < \tau^\tau$$

to obtain(6.43). For $c=0$, the author writes that $\binom{a+l-d}{0}=0$ regardless of $a+l-d$ in(6.48), and the same analysis gives(6.44). (3) Each error matrix has dimension $(\tau+1)^2$ and entries absolutely $<\tau^{1.5\tau}/n$ by(2). □

Remark 6.25. The specialty of $c=0$ in(6.44) is that $\binom{|A|+l-d}{0}$ is always 1. For $c>0$, $\binom{|A|+l-d}{c}$ can be zero or negative depending on the order between $0, c, |A|+l-d$, so its expected matrix may not be PSD.

Definition 6.26. For $m,t\in\mathbb{N}$, the factorial Hankel matrix is $H_{m,t}(i,j) = (i+j+t)!$ for $0\le i,j\le m$ (6.49).

Proposition 6.27 (Almost common decomposition). (1)

$$H_{m,t} = L_m[N_{m,t}D_{m,t}(N_{m,t})^\top]L_m^\top,$$

where $L_m, D_{m,t}$ are diagonal and $N_{m,t}$ lower-triangular,

$$L_m(i,i) = i!,\quad D_{m,t}(i,i) = \prod_{t'=1}^{t}(i+t'),\quad N_{m,t}(i,j) = \binom{i+t}{i-j}.$$

L_m is independent of t and $H_{m,t}$ positive. (2) Let J_m be the $(m+1)\times(m+1)$ lower-triangular Jordan block, equal to 1 if $i=j$ or $i=j+1$ and 0 otherwise. Then $N_{m,t+1}=N_{m,t}J_m$ (6.50).

Proof. This follows from a direct inspection. □

Proposition 6.28. If $t+1 > 8\max\{r^2, m\}$ (6.51), then $H_{m,t+1} \succeq 2r^2 H_{m,t}$.

Proof. By Proposition6.27 it suffices to show $J_m D_{m,t+1} J_m^\top \succeq 2r^2 D_{m,t}$. For $x^\top = (x_0, \dots, x_m)$ compare

$$q_{t+1}(x) := (x^\top J_m) D_{m,t+1} (J_m^\top x), \qquad q_t(x) := 2r^2 x^\top D_{m,t} x. \tag{6.52}$$

Put $\alpha(y) = 2r^2 \prod_{t'=1}^{t}(y+t')$, $\beta(y) = \prod_{t'=1}^{t+1}(y+t')$. Then $q_{t+1}(x) = \sum_{i=0}^{m} \beta(i)(x_i + x_{i+1})^2$, $x_{m+1} := 0$, while $q_t(x) = \sum_{i=0}^{m} \alpha(i) x_i^2$. Completing squares gives

$$q_{t+1}(x) = \sum_{i=0}^{m} \left[\alpha(i) x_i^2 + (\beta(i) - \alpha(i)) \left(x_i + \frac{\beta(i)}{\beta(i) - \alpha(i)} x_{i+1} \right)^2 - \frac{\beta(i)^2}{\beta(i) - \alpha(i)} x_{i+1}^2 \right].$$

Define $b_0 = 1 - \alpha(0)/\beta(0)$ and

$$b_i = 1 - \frac{\alpha(i)}{\beta(i)} - \frac{\beta(i-1)}{\beta(i)} \frac{1}{b_{i-1}}. \tag{6.53}$$

Then

$$q_{t+1}(x) = \underbrace{\sum_{i=0}^{m} \alpha(i) x_i^2}_{q_t(x)} + \sum_{i=0}^{m} \beta(i) b_i \left(x_i + \frac{1}{b_i} x_{i+1} \right)^2. \tag{6.54}$$

Claim 6.29. Each $b_i > 1/2$ for $i \le m$.

Proof of claim. $b_0 = 1 - 2r^2/(t+1)$ and, for $i \ge 1$,

$$b_i = 1 - \frac{2r^2}{t+1+i} - \frac{i}{t+1+i} \frac{1}{b_{i-1}}. \tag{6.55}$$

Induct: $b_0 > 1/2$ by(6.51), and

$$b_i \ge 1 - \frac{2r^2}{t+1} - \frac{m}{t+1} \cdot 2 > 1/2$$

by(6.51) and the inductive hypothesis. This proves the claim. Since b_i are positive, (6.54) gives $q_{t+1} \ge q_t$, proving the proposition. □

Apply this to(6.47): on its support,

$$P_{|A|+l} = \frac{1}{(8\tau^2)!} H_{\tau - (d/2 - |A|),\ d - |A| + 8\tau^2 + l}.$$

Corollary 6.30 (Positiveness of $\mathbb{E}[Q_{0,0}^R]$). In(6.45), regarding matrices on their support,

$$\sum_{l=0}^{r} (-1)^{r-l} \frac{\binom{r}{l}}{l!} P_{|A|+l} + E_A^R \succ \operatorname{diag}(\tau^{-6\tau})_{0 \le i \le \tau - (d/2 - |A|)}. \tag{6.56}$$

In particular,

$$\mathbb{E}[Q_{0,0}^R] \succ \sum_{A \supseteq R} 1_{A,A} \otimes \operatorname{diag}(\tau^{-6\tau})_{d/2 - |A| \le i \le \tau} = \operatorname{diag}(\tau^{-6\tau})_{\mathcal{S}^R \times \mathcal{S}^R}. \tag{6.57}$$

Proof. The last part follows by checking support and that tensoring with a nonzero PSD matrix preserves $\succ$. For(6.56), fix A and put

$$\tau_0 = \tau - (d/2 - |A|), \qquad t_0 = d - |A| + 8\tau^2. \tag{6.58}$$

Then

$$\sum_{l=0}^{r} (-1)^{r-l} \frac{\binom{r}{l}}{l!} P_{|A|+l} = \frac{1}{(8\tau^2)!}(X_r + X_{r-2} + \cdots), \tag{6.59}$$

where for $0 \le v \le \lfloor r/2 \rfloor$,

$$X_{r-2v} = \frac{\binom{r}{r-2v}}{(r-2v)!}\left(H_{\tau_0,t_0+r-2v} - \frac{(r-2v)^2}{2v+1} H_{\tau_0,t_0+r-2v-1}\right), \quad H_{\tau_0,-1} := 0.$$

The second coefficient is $\le r^2$. As $t_0 > 8\max\{r^2, \tau_0\}$, Proposition6.28 gives

$$X_{r-2v} \succeq \frac{\binom{r}{r-2v}}{(r-2v)!} \max\{\tfrac{1}{2} H_{\tau_0,t_0+r-2v},\ r^2 H_{\tau_0,t_0+r-2v-1}\}.$$

In particular,

$$\begin{aligned} \sum_{l=0}^{r} (-1)^{r-l} \frac{\binom{r}{l}}{l!} P_{|A|+l} &\succeq \frac{1}{(8\tau^2)!} H_{\tau_0,t_0} \\ &= L[N_{t_0} \tfrac{D_{t_0}}{(8\tau^2)!} (N_{t_0})^\top] L^\top, \end{aligned} \tag{6.60}$$

omitting index τ_0 on the RHS. Claim6.31 below implies the RHS is $\succ L\,\mathrm{diag}(\tau^{-5\tau})L \succeq \mathrm{diag}(\tau^{-5\tau})$. By Proposition6.24(3), $\|E_A^R\| < \tau^{2\tau}/n < \tau^{-6\tau}$ in the parameter regime. Thus the LHS of(6.56) is $\succeq \mathrm{diag}(\tau^{-5\tau} - \tau^{-6\tau}) \succeq$ its RHS. $\square$

Claim 6.31. In Corollary6.30's notation,

$$N_{t_0}^{-1}(i,j) = (-1)^{i-j}\binom{i+t_0}{i-j}, \quad 0 \le i,j \le \tau_0, \tag{6.61}$$

and

$$N_{t_0} \frac{D_{t_0}}{(8\tau^2)!} (N_{t_0})^\top \succ \mathrm{diag}(\tau^{-5\tau})_{0 \le i \le \tau_0}. \tag{6.62}$$

Proof. For(6.61), multiplying by N_{t_0} gives at (i,j)

$$\sum_{j \le k \le i} (-1)^{i-k} \binom{i+t_0}{i-k}\binom{k+t_0}{k-j} = \sum_{k'=0}^{i'} (-1)^{i'-k'} \binom{i'+j+t_0}{i'-k'}\binom{k'+j+t_0}{k'},$$

where $i' = i-j$, $k' = k-j$. To see identity, let $D_m[(1+x)^a]$ be the coefficient of x^m for $m \ge 0, a \in \mathbb{Z}$. The RHS is

$$\begin{aligned} &(-1)^{i'} \sum_{k'=0}^{i'} D_{i'-k'}[(1+x)^{i'+j+t_0}] D_{k'}[(1+x)^{-(t_0+j+1)}] \\ &= (-1)^{i'} D_{i'}[(1+x)^{i'+j+t_0-(t_0+j+1)}] \\ &= (-1)^{i'} D_{i'}[(1+x)^{i'-1}] = \mathbf{1}_{i'=0}. \end{aligned}$$

For(6.62), equivalently

$$\frac{D_{t_0}}{(8\tau^2)!} \succ N_{t_0}^{-1}\tau^{-5\tau}(N_{t_0}^{-1})^\top. \tag{6.63}$$

Set $a_0 = \tau^{-5\tau}$. The RHS quadratic form is

$$x^\top N_{t_0}^{-1} a_0 (N_{t_0}^{-1})^\top x = a_0 \sum_{j=0}^{\tau_0} y_j^2, \tag{6.64}$$

with $y_j = (x^\top N_{t_0}^{-1})_j = \sum_{i=j}^{\tau_0} (-1)^{i-j}\binom{i+t_0}{i-j} x_i$. By Cauchy–Schwarz, $y_j^2 \le \tau_0 \sum_{i=j}^{\tau_0} \binom{i+t_0}{i-j}^2 x_i^2$. Hence

$$\begin{aligned} a_0 \sum_j y_j^2 &\le a_0 \sum_{i=0}^{\tau_0} x_i^2 \left(\tau_0 \sum_{j=0}^{i} \binom{i+t_0}{i-j}^2 \right) \\ &< \sum_{i=0}^{\tau_0} \tau^{-5\tau} (9\tau^2)^{2i+2} x_i^2. \end{aligned}$$

But the LHS diagonal of(6.63) satisfies

$$\frac{D_{t_0}(i,i)}{(8\tau^2)!} \ge (8\tau^2)^{-(d/2-|A|)} > \tau^{-2d} > \tau^{-5\tau}(9\tau^2)^{2i+2},$$

using $i \le \tau_0 < \tau$ and $d \ll \tau$. Thus(6.63) holds. □

Corollary 6.32 (Positive structural part).

$$D_{\mathrm{Cl}}^\tau \mathbb{E}[Q_{0,0}^R] D_{\mathrm{Cl}}^\tau \succeq \tau^{-6\tau} \operatorname{diag}(\widetilde{\mathrm{Cl}})_{\mathcal{S}^R \times \mathcal{S}^R}.$$

Proof. This follows from Corollary6.30 and $D_{\mathrm{Cl}}^2(A,A) = \widetilde{\mathrm{Cl}}(A)$ in Definition6.22. □

Author Section 6.6: bounds on the remaining matrices

We bound the pseudorandom part of $Q_{0,0}^R$, all $Q_{0,k}^R$ for $k > 0$, and all $\omega^{-c} Q_{c,k}^R$ for $c > 0$, by Lemmas6.34,6.37,6.38. This proves Lemma6.21. Care is required on the block structure and supports.

Definition 6.33 (0-1 diagonal-clique matrix). D' is the 0-1 version of D_{Cl}^τ, diagonal with $D'((A,i),(A,i)) = \mathrm{Cl}_A$ for all $A \in \binom{[n]}{\le d/2}$, $0 \le i \le \tau$.

Lemma 6.34. W.p. $> 1 - n^{-9\log n}$, for all $R \in \binom{[n]}{\le d/2}$,

$$\pm(Q_{0,0}^R - D_{\mathrm{Cl}}^\tau \mathbb{E}[Q_{0,0}^R] D_{\mathrm{Cl}}^\tau)(G) \preceq n^{-\epsilon} \operatorname{diag}(\widetilde{\mathrm{Cl}}(G))_{\mathcal{S}^R \times \mathcal{S}^R}. \tag{6.65}$$

Proof. Fix R and abbreviate $Q_{\mathrm{ps}} := Q^R_{0,0} - D^\tau_{\mathrm{Cl}}\mathbb{E}[Q^R_{0,0}]D^\tau_{\mathrm{Cl}} = (Q_{\mathrm{ps},(i,j)})_{0\le i,j\le\tau}$. Block (i,j) is supported on $S^2_{i,j}$, $S_{i,j} = \{A : |A| + \min\{i,j\} \ge d/2\}$. For $A \ne B$,

$$\begin{aligned} Q_{\mathrm{ps},(i,j)}(A,B) &= Q^R_{0,0}((A,i),(B,j)) \\ &= \sum_{\substack{T_m : |V(T_m)\cup A\cup B|\le\tau \\ A,B\in \mathrm{mSep}_{A,B}(T_m)}} (\omega/n)^{|V(T_m)\cup A\cup B| - (|A|+|B|)/2} q(A,B;T_m)\chi_{T_m}. \end{aligned} \tag{6.66}$$

On the diagonal,

$$Q_{\mathrm{ps},(i,j)}(A,A) = \sum_{T_m : 1\le |V(T_m)\setminus A|\le \tau - |A|} (\omega/n)^{|V(T_m)\cup A| - |A|} q(A,A;T_m)\chi_{T_m}. \tag{6.67}$$

Abbreviate $q(A,B;T_m) := Y_0(|R|, |V(T_m)\cup A\cup B| + (i+j))$. Two properties:

$$q(A,B;T_m) \text{ depends only on } |V(T_m)\cup A\cup B| \text{ for fixed } (A,B), \tag{6.68}$$

$$q(A,B;T_m) < \tau^{5\tau} \quad \text{by Lemma6.8(4).} \tag{6.69}$$

Thus each entry factors through $\mathrm{Cl}_{A\cup B}$, hence $\mathrm{Cl}_A\mathrm{Cl}_B$. In particular,

$$Q_{\mathrm{ps}} = D' Q_{\mathrm{ps}} D'. \tag{6.70}$$

Claim 6.35. W.p. $> 1 - n^{-9.5\log n}$, all blocks satisfy

$$\pm Q_{\mathrm{ps},(i,j)} \prec n^{-1.1\epsilon}\,\mathrm{diag}(2^{\binom{|A|}{2}})_{S^R_{\min\{i,j\}}\times S^R_{\min\{i,j\}}},$$

where $S^R_a = \{A \in \binom{[n]}{\le d/2} : |A| + a \ge d/2\}$. The lemma follows from this claim and(6.70). For $b \in [0, d/2]$, let $I_b := \{i : d/2 - b \le i \le \tau\}$, $\mathcal{P}^R_b := \{A \subseteq [n] : R \subseteq A, |A| = b\}$, and $Q_{\mathrm{ps};b}$ be the principal minor on $W_b := (\mathcal{P}^R_b \times I_b)^2$, zero elsewhere. These W_b partition the support where $|A| = |B|$. Remark6.16(2) gives

$$Q_{\mathrm{ps}} = \sum_{b=0}^{d/2} Q_{\mathrm{ps};b}. \tag{6.71}$$

Each is blockwise on I_b^2, blocks principal minors of $Q_{\mathrm{ps},(i,j)}$. By Claim6.35 w.p. $> 1 - n^{-9.5\log n}$ each signed block is $\prec n^{-1.5\epsilon}\,\mathrm{diag}(2^{\binom{b}{2}})$, so $\pm Q_{\mathrm{ps};b} \prec \tau^2 n^{-1.5\epsilon}\,\mathrm{diag}(2^{\binom{b}{2}})_{W_b} \prec n^{-\epsilon}\,\mathrm{diag}(2^{\binom{b}{2}})_{W_b}$. By(6.71) and union-bound over b, $\pm Q_{\mathrm{ps}} \prec n^{-\epsilon}\,\mathrm{diag}(2^{\binom{|A|}{2}})_{\mathcal{S}^R\times\mathcal{S}^R}$ w.p. $1 - n^{-9\log n}$. Insert this into(6.70), with $\widetilde{\mathrm{Cl}}_A = 2^{\binom{|A|}{2}}\mathrm{Cl}_A$ and $\mathrm{Cl}^2_A = \mathrm{Cl}_A$, obtaining(6.65). □

Proof of Claim 6.35. Use Section4's norm bounds. Fix (i,j), split $Q_{\mathrm{ps},(i,j)}$ into diagonal and off-diagonal parts.

Diagonal part. For supported (A,A), (6.67) gives

$$Q^{\mathrm{diag}}_{\mathrm{ps},(i,j)}(A,A) = \widetilde{\mathrm{Cl}}_A \underbrace{\sum_{\substack{T_m : 1\le |V(T_m)\setminus A|\le\tau-|A| \\ T_m\cap E[A]=\emptyset}} (\omega/n)^{|V(T_m)\setminus A|} q(A,A;T_m)\chi_{T_m}}_{=:g(A)}.$$

For fixed A, q depends only on $|V(T_m)\setminus A|$, denote it $q(v)$. For $1\le v\le \tau-|A|$, let $U_1^v,\dots,U_{h(v)}^v$ be all shapes $(A,A;T)$ with $T\cap E[A]=\emptyset$, $|V(T)\setminus A|=v$. Clearly,

$$h(v)\le 2^{|A|v+v^2}. \tag{6.72}$$

W.p. $>1-n^{-9.6\log n}$,

$$\begin{aligned}
|g(A)| &= \left|\sum_{v=1}^{\tau-|A|}(\omega/n)^v q(v)\sum_{x=1}^{h(v)}\sum_{T_m:(A,A;T_m)\text{ has shape }U_x^v}\chi_{T_m}\right| \\
&\le \sum_{v=1}^{\tau-|A|}(\omega/n)^v q(v)\sum_{x=1}^{h(v)}\|M_{U_x^v}\| \\
&\le \sum_{v=1}^{\tau-|A|}(\omega/n)^v \tau^{5\tau}\sum_{x=1}^{h(v)}\|M_{U_x^v}\| \\
&< \sum_{v=1}^{\tau}(\omega/n)^v\tau^{5\tau}2^{|A|v+v^2}n^{v/2}2^{O(|A|+v)} \\
&< \sum_{v=1}^{\tau} n^{-3\epsilon v}n^{\epsilon v} < n^{-1.2\epsilon}.
\end{aligned}$$

The character sum equals $M_{U_x^v}(A,A)$ by Definition4.7; these matrices are diagonal. Use(6.69),(6.72),Theorem4.8 and the parameter regime in the respective inequalities.

Off-diagonal part. By coefficient symmetry(6.68), it is a sum of graphical matrices. Let $U_1^{s,t},\dots,U_{h(s,t)}^{s,t}$ be all distinct shapes $(A,B;T)$ with $|A|=|B|=s$, $A\ne B$, $A,B\in \mathrm{mSep}_{A,B}(T)$ and $|V(T)\cup A\cup B|=t$. By(6.66), it is block-diagonal by $s=d/2-i,\dots,d/2$, with block

$$Q^{\mathrm{off}}_{\mathrm{ps},(i,j)}(s)=\sum_{s<t\le\tau}(\omega/n)^{t-s}\sum_{x=1}^{h(s,t)}q(U_x^{s,t})M_{U_x^{s,t}}.$$

By Theorem4.8 w.p. $>1-n^{-9.8\log n}$,

$$\|Q^{\mathrm{off}}_{\mathrm{ps},(i,j)}(s)\|\le\sum_{s<t\le\tau}(\omega/n)^{t-s}h(t,s)n^{(t-s)/2}2^{O(t)}(\log n)^{O(t-s)}. \tag{6.73}$$

Clearly $h(t,s)\le 2^{\binom{t}{2}+O(t)}$. With the same probability,

$$\begin{aligned}
\text{RHS of(6.73)} &\le \sum_{\substack{d/2-\max\{i,j\}\le s\le d/2\\ s<t\le\tau}}(\omega/n)^{t-s}2^{\binom{t}{2}+O(t)}n^{(t-s)/2}(\log n)^{O(t-s)} \\
&< \sum_{\substack{d/2-\max\{i,j\}\le s\le d/2\\ s<t\le\tau}} n^{-2\epsilon(t-s)}2^{O(t)}2^{\binom{s}{2}}(2^{t+s}\log n)^{O(t-s)} \\
&< 2^{\binom{s}{2}}n^{-1.9\epsilon},
\end{aligned}$$

by the parameter regime. Add these diagonal blocks to obtain $\pm Q^{\mathrm{off}}_{\mathrm{ps},(i,j)} \prec n^{-1.9\epsilon}\operatorname{diag}(2^{\binom{|A|}{2}})$ on the stated support. Finally union-bound: w.p. $> 1 - n^{-9.5\log n}$,

$$\pm Q_{\mathrm{ps},(i,j)} = \pm(Q^{\mathrm{diag}}_{\mathrm{ps},(i,j)} + Q^{\mathrm{off}}_{\mathrm{ps},(i,j)}) \prec n^{-1.5\epsilon}\operatorname{diag}(2^{\binom{|A|}{2}})_{S^R_{\min\{i,j\}}\times S^R_{\min\{i,j\}}}.$$

This completes the claim. □

Corollary 6.36 (Positiveness of $Q^R_{0,0}$). For every $R \in \binom{[n]}{\le d/2}$, w.p. $> 1 - n^{-8\log n}$,

$$Q^R_{0,0}(G) \succeq \tau^{-6.1\tau}\operatorname{diag}(\widetilde{\mathrm{Cl}}(G))_{\mathcal{S}^R\times\mathcal{S}^R}.$$

Proof. By Lemma6.34 and Corollary6.32, where $\tau^{-6.1\tau} \gg n^{-\epsilon/10}$ in our parameter regime. □

Author remaining block bounds and final PSD proof

Lemma 6.37 (Bounds on $Q^R_{0,k}$). W.p. $> 1 - n^{-9\log n}$, all $R \in \binom{[n]}{\le d/2}$ and $1 \le k \le d/2$ satisfy

$$\pm Q^R_{0,k}(G) \preceq n^{-k/10}\operatorname{diag}(\widetilde{\mathrm{Cl}}(G))_{\mathcal{S}^R\times\mathcal{S}^R}.$$

Proof. Union-bound over (R,k), fixing them first, and abbreviate $Q = Q^R_{0,k}$, $q_k = q^R_{0,k}$. From Lemma6.19(3), Q is supported on $\mathcal{S}^R \times \mathcal{S}^R$ and

$$Q((A,i),(B,j)) = \sum_{T_m : |V(T_m)\cup A\cup B| \le \tau} (\omega/n)^{t-s} q_k(R_m,i,j)\chi_{T_m}, \tag{6.74}$$

where $t = |A\cup B|$, $s = (|A|+|B|)/2$. For fixed (i,j),

$$q_k(\cdot,i,j) \text{ is symmetric w.r.t. shapes,} \tag{6.75}$$

$$|q_k(R_m,i,j)| \le \tau^{5\tau}(\omega/n^{1-\epsilon})^{s-p+k/3}, \tag{6.76}$$

where p is the max number of vertex-disjoint A–B paths. Symmetry makes entries factor through $\mathrm{Cl}(A)\mathrm{Cl}(B)$, so

$$Q = D'QD'. \tag{6.77}$$

It suffices that, w.p. $> 1 - n^{-9.5\log n}$,

$$\pm Q \prec n^{-k/10}\operatorname{diag}(2^{\binom{|A|}{2}})_{\mathcal{S}^R\times\mathcal{S}^R}. \tag{6.78}$$

Insert this into(6.77) as in Lemma6.34. To prove(6.78), express blocks as graphical matrices. For fixed (i,j) and $(s_1,s_2) \in \{0,\dots,d/2\}^2$ with $s_1+i, s_2+j \ge d/2$, $t \ge \max\{s_1,s_2\}$, let $U^{t;s_1,s_2}_x$, $1 \le x \le h(t;s_1,s_2)$, be all shapes with $|A| = s_1, |B| = s_2$, total size t. By(6.74) and symmetry,

$$Q_{(i,j)} = \sum_{\substack{(t;s_1,s_2):\ s_1+i,s_2+j\ge d/2 \\ \tau\ge t\ge s_1,s_2}} \sum_{x=1}^{h(t;s_1,s_2)} q_k(U^{t;s_1,s_2}_x,i,j) M_{U^{t;s_1,s_2}_x}.$$

View this blockwise by (s_1, s_2):

$$Q_{(i,j)} = \sum_{s_1,s_2:\ s_1+i,s_2+j\geq d/2} Q_{(s_1,i),(s_2,j)}, \tag{6.79}$$

where

$$Q_{(s_1,i),(s_2,j)} := \sum_{s_1,s_2\leq t\leq \tau} \sum_{x=1}^{h(t;s_1,s_2)} q_k(U_x^{t;s_1,s_2}, i, j) M_{U_x^{t;s_1,s_2}}. \tag{6.80}$$

This is a $\binom{[n]}{s_1} \times \binom{[n]}{s_2}$ matrix in block(i,j). By Theorem4.8 and(6.76), w.p. $> 1 - n^{-10\log n}$,

$$\|Q_{(s_1,i),(s_2,j)}\| \leq \sum_{\substack{t\leq\tau \\ t\geq s_1,s_2}} h(t;s_1,s_2)(\omega/n)^{t-s}(\omega/n^{1-\epsilon})^{s-p+k/3} \cdot n^{(t-p)/2} 2^{O(t)} (\log n)^{O(t-s)}, \tag{6.81}$$

where $s = (s_1+s_2)/2$ and p is the max path count in the shape. Since $h(t;s_1,s_2) \leq 2^{\binom{t}{2}+O(t)} = 2^{\binom{s}{2}+O(t)+(t+s)(t-s)}$, and $k > 0$, $2^{O(t)} < n^{\epsilon/10}$, $\tau^{5\tau} < n^{1/30}$, the RHS is

$$< 2^{\binom{s}{2}} \tau^{5\tau} n^{-k/6} n^{-\epsilon(t-s)} < 2^{\binom{s}{2}} n^{-k/8}. \tag{6.82}$$

Finally extend double-blocks by zero to $\mathcal{S}^R \times \mathcal{S}^R$ and sum:

$$Q = \sum_{(s_1,i),(s_2,j):s_1+i,s_2+j\geq d/2} Q_{(s_1,i),(s_2,j)}. \tag{6.83}$$

For each summand, (6.82) and Cauchy–Schwarz give

$$\pm Q_{(s_1,i),(s_2,j)} \prec \frac{n^{-k/8}}{2} \left(2^{\binom{s_1}{2}} \mathrm{Id}_{(s_1,i),(s_1,i)} + 2^{\binom{s_2}{2}} \mathrm{Id}_{(s_2,j),(s_2,j)}\right).$$

Thus w.p. $> 1 - n^{-9.5\log n}$,

$$\pm Q \prec \tau^2 n^{-k/8} \operatorname{diag}(2^{\binom{|A|}{2}}) \prec n^{-k/10} \operatorname{diag}(2^{\binom{|A|}{2}})$$

on $\mathcal{S}^R \times \mathcal{S}^R$, proving(6.78). □

Lemma 6.38 (Bounds for $c > 0$). W.p. $> 1 - n^{-9\log n}$, for every $R \in \binom{[n]}{\leq d/2}$, $0 < c \leq |R|$, $0 \leq k \leq d/2$,

$$\pm \omega^{-c} Q_{c,k}^R \preceq n^{-c/3} \operatorname{diag}(\widetilde{\mathrm{Cl}})_{\mathcal{S}^R\times\mathcal{S}^R}. \tag{6.84}$$

Proof. Union-bound over (R, c, k): it suffices to prove the inequality for a fixed triple with probability $> 1 - n^{-9.5\log n}$. If $k > 0$, the proof is identical to Lemma6.37: by Lemma6.19 the same symmetry and coefficient-size

conditions(6.75),(6.76) hold, with the same support. So consider $c > 0, k = 0$. Definition6.15 gives

$$Q^R_{c,0}((A,i),(B,j)) = \sum_{\substack{T_m : |V(T_m)\cup A\cup B| \le \tau \\ A,B \in \mathrm{mSep}_{A,B}(T_m)}} (\omega/n)^{|V(T_m)\cup A\cup B| - (|A|+|B|)/2} \cdot Y_c(|R|, |V(T_m)\cup A\cup B| + (i+j))\chi_{T_m}. \tag{6.85}$$

Lemma6.8(4) bounds Y_c by $\tau^{5\tau}$. Put $t = |V(T_m)\cup A\cup B|$, $s = (|A|+|B|)/2 = |A| = |B|$. The coefficient is bounded by $(\omega/n)^{t-s}\tau^{5\tau}$. Support and symmetry let the proof proceed as before to(6.81), now with a single term

$$h(t; s_1, s_2)(\omega/n)^{t-s}\tau^{5\tau} n^{(t-p)/2} 2^{O(t)} (\log n)^{O(t-s)}.$$

Every appearing ribbon has $A, B \in \mathrm{mSep}_{A,B}(T_m)$, so $p = s$ (the specialty of $k = 0$). Replace(6.82)'s bound by $\tau^3 2^{\binom{s}{2}} n^{-3\epsilon(t-s)}\tau^{5\tau} 2^{O(t)} < 2^{\binom{s}{2}}\tau^{6\tau}$ and proceed to the last line as before, obtaining

$$\pm Q^R_{c,0} \prec \tau^{7\tau}\,\mathrm{diag}(2^{\binom{|A|}{2}})_{\mathcal{S}^R\times\mathcal{S}^R}.$$

Since $c \ge 1$, $\omega = n^{1/2-4\epsilon}$ with $\epsilon < 1/40$, and $\tau^{7\tau} < n^{1/15}$, parameters give

$$\pm\omega^{-c} Q^R_{c,0} \prec n^{-c/3}\,\mathrm{diag}(2^{\binom{|A|}{2}}).$$

Using again $Q^R_{c,0} = D' Q^R_{c,0} D'$ converts this to the normalized clique diagonal in(6.84). □

Lemma6.21 now follows immediately from Corollary6.36 and Lemmas6.37/6.38.

Author Section 6.7: last step

Recall $\mathcal{P}^R = \{I \in \binom{[n]}{d/2} : R \subseteq I\}$. Main Lemma6.9 requires, w.p. $1 - n^{-5\log n}$ simultaneously,

$$M^R_0 \succeq n^{-d}\,\mathrm{diag}(\widetilde{\mathrm{Cl}})_{\mathcal{P}^R\times\mathcal{P}^R}, \tag{6.86}$$

$$\pm\omega^{-c} M^R_c \preceq n^{-c/6} M^R_0 \quad (0 < c \le |R|). \tag{6.87}$$

Lemma 6.39. For every $R \in \binom{[n]}{\le d/2}$, on every G,

$$\widetilde{L}_R D^\tau\,\mathrm{diag}(\widetilde{\mathrm{Cl}})_{\mathcal{S}^R\times\mathcal{S}^R} D^\tau (\widetilde{L}_R)^\top \succeq (\omega/n)^{d/2}\,\mathrm{diag}(\widetilde{\mathrm{Cl}})_{\mathcal{P}^R\times\mathcal{P}^R}.$$

Proof. Fix R. Regard supports on $[n'] = [n] - R$ and reindex matrices by $\binom{[n']}{d'/2}$ or $\binom{[n']}{\le d'/2}$ with $d'/2 = d/2 - |R|$; τ is unchanged. Still write $\widetilde{\mathrm{Cl}}(X)$ for $\widetilde{\mathrm{Cl}}(X\sqcup R)$ for $X \subseteq [n']$. Since $D^\tau\,\mathrm{diag}(\widetilde{\mathrm{Cl}})D^\tau$ is nonnegative and diagonal,

$$\widetilde{L}_R[D^\tau\,\mathrm{diag}(\widetilde{\mathrm{Cl}})D^\tau](\widetilde{L}_R)^\top \succeq L_{R,0}[D^\tau\,\mathrm{diag}(\widetilde{\mathrm{Cl}})D^\tau](L_{R,0})^\top. \tag{6.88}$$

Recall $\widetilde{L}_R = (L_{R,0}, \dots, L_{R,\tau})$. Further decompose $L_{R,0} = (L^0_{R,0}, \dots, L^{d'/2}_{R,0})$ by the size of the column set. In particular,

$$L^{d/2-|R|}_{R,0} = (0, \dots, 0, \operatorname{diag}(\widetilde{\mathrm{Cl}})_{\binom{[n']}{d'/2}\times\binom{[n']}{d'/2}}),$$

since Definition6.11 allows only zero-reduced-size ribbons, which with its other conditions means $A = I$ and $T' \subseteq E(I)$. Thus the RHS of(6.88) is $\succ (\omega/n)^{d/2} \operatorname{diag}(\widetilde{\mathrm{Cl}})$ on that minor. Translating back gives the lemma. □

Proof of Main Lemma 6.9. Fix R. By Lemma6.19, for each $c \le |R|$,

$$M^R_c = \widetilde{L}_R[D^\tau(Q^R_{c,0} - Q^R_{c,1} + \cdots \pm Q^R_{c,d})D^\tau](\widetilde{L}_R)^\top + E^R_c. \tag{6.89}$$

The following bounds each hold with probability $> 1 - n^{-8\log n}$, and union-bound makes the overall probability $> 1 - n^{-5\log n}$. For(6.86),

$$\begin{aligned}
M^R_0 &\succeq \tau^{-7\tau}\widetilde{L}_R D^\tau \operatorname{diag}(\widetilde{\mathrm{Cl}})D^\tau(\widetilde{L}_R)^\top + E^R_0 \\
&\qquad \text{Lemma6.21(1)} \\
&\succeq \tau^{-7\tau}(\omega/n)^{d/2} \operatorname{diag}(\widetilde{\mathrm{Cl}})_{\mathcal{P}^R\times\mathcal{P}^R} + E^R_0 \\
&\qquad \text{Lemma6.39} \\
&\succeq (\tau^{-7\tau}(\omega/n)^{d/2} - n^{-\epsilon\tau/2}) \operatorname{diag}(\widetilde{\mathrm{Cl}})_{\mathcal{P}^R\times\mathcal{P}^R} \\
&\qquad \text{Lemma6.19(4)} \\
&\succeq n^{-d} \operatorname{diag}(\widetilde{\mathrm{Cl}})_{\mathcal{P}^R\times\mathcal{P}^R} \\
&\qquad \text{parameter regime.}
\end{aligned}$$

For(6.87), fix $1 \le c \le |R|$:

$$\begin{aligned}
M^R_c &\preceq \omega^c n^{-c/4}\widetilde{L}_R D^\tau \operatorname{diag}(\widetilde{\mathrm{Cl}})D^\tau(\widetilde{L}_R)^\top + E^R_c \\
&\preceq \omega^c n^{-c/4}\tau^{7\tau}(M^R_0 - E^R_0) + E^R_c \\
&\preceq \omega^c n^{-c/5} M^R_0 + (\omega^c n^{-c/5} + 1)n^{-\epsilon\tau/2} \operatorname{diag}(\mathrm{Cl})_{\mathcal{P}^R\times\mathcal{P}^R}.
\end{aligned}$$

The respective steps use Lemma6.21(2), then Lemma6.21(1) and(6.89), then Lemma6.19(4). So

$$\begin{aligned}
\omega^{-c}M^R_c &\preceq n^{-c/5}M^R_0 + 2n^{-\epsilon\tau/2} \operatorname{diag}(\mathrm{Cl})_{\mathcal{P}^R\times\mathcal{P}^R} \\
&\preceq (n^{-c/5} + 2n^d n^{-\epsilon\tau/2})M^R_0 \\
&\qquad \text{by(6.86), } \widetilde{\mathrm{Cl}} \ge \mathrm{Cl} \\
&\preceq n^{-c/6}M^R_0 \\
&\qquad \text{since } c \le |R| \le d/2 \text{ and parameters.}
\end{aligned}$$

The same analysis holds for $-\omega^{-c}M^R_c$. This proves Main Lemma6.9, hence Theorem6.1 and the main exact-clique lower bound as shown above. □

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